

PHYSICAL GEODESY

BY

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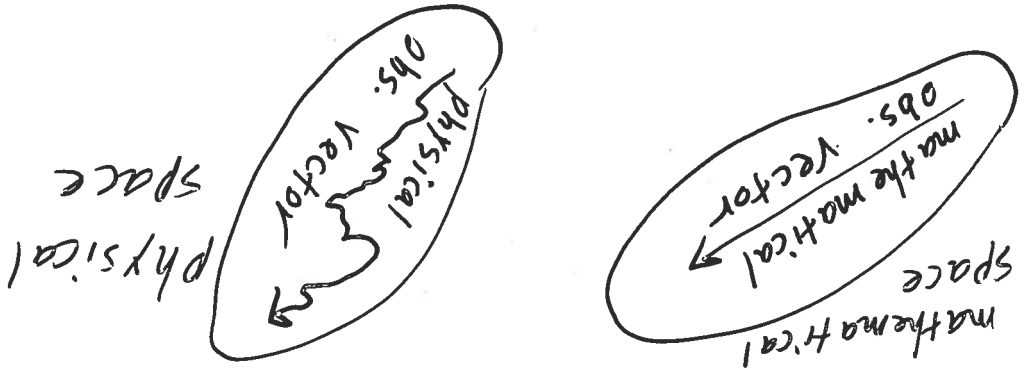
Physical Geodesy

Contents:

- 1- Introduction (physical space-mathematical space)
- 2- Attraction and potential of gravitation, Newton's formula, Poisson's Equation for the Earth's potential of gravitation
- 3- Laplace Equation, harmonic functions, Dirichlet's principle, Boundary value problems, Laplace boundary value problem, spherical harmonics
- 4- Real and normal gravity fields, the anomalies gravity field, geoidal height, deflections of the vertical, spherical harmonic presentation of the Earth's gravity field parameters
- 5- Gravimetric methods of geoidal height determination, Stokes's formula, Bruns's formula, Formula of Venig-Meinesz
- 6- Gravimetry, gravity measurements, Gravimeters, Gravity Reductions
- 7- Heights, Geopotential numbers,
- 8- Non-gravimetric method of geoidal-height determination

Introduction:

Two Important surfaces which are used in geodesy are: physical surface of the Earth and mathematical surface of the Earth. The observations are being done in the physical surface, but the computations are being done in the mathematical surface of the Earth. $\Rightarrow$ Geometric geodesy physical geodesy



Biaxial ellipsoid is used as the figure of the Earth in Geodesy.

In this course we are going to understand better the physical space of the Earth and also to obtain the parameters which relate the two physical and mathematical spaces (the most important one geoidal height and deflection of vertical)

Physical effects of the Earth on observations?

- Refraction
- Coriolis force
- Earth's gravity ^② important
- magnetic force

gravity field:

According to Newton's law of gravitation a body of mass M attracts body of mass m by a force of $\vec{F}$ whose magnitude is proportional to the product of the two masses and inversely proportional to the square of their distance Dr :

$$F = G \frac{Mm}{Dr^2}$$

①

This force has become known as the gravitational force and is also called gravitational attraction, or

Newton's attraction. the constant of G is known as Newton's gravitational constant. It can be

interpreted as the general property of any mass.

$$G = 6.672 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2}$$

Although the masses M, m attract each other in a completely symmetrical way, it is convenient to call one of them the attracting mass and the other the attracted mass. For simplicity we set the attracting mass by M . the formula

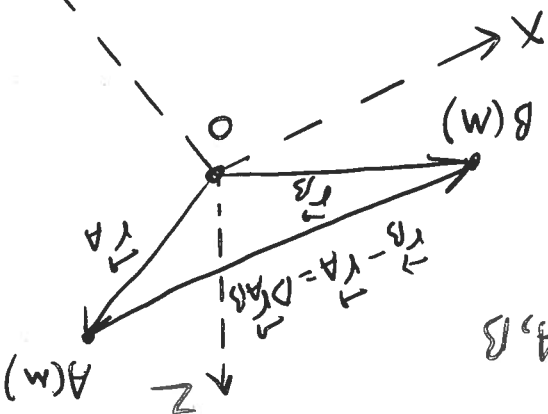
$$F = G \frac{M}{Dr^2}$$

②

expresses the force exerted by the mass M on a

③

unit mass located a distance r from M .



Taking two physical bodies A, B

with masses m and M , and

considering their dimensions

negligibly small compared

with their distance,

the following vector equation can be written for

the gravitational force that B exerts on A:

$$\vec{F}_{B \rightarrow A} = G \frac{M m}{|\vec{r}_B - \vec{r}_A|^3} (\vec{r}_B - \vec{r}_A) \quad (3) \quad \vec{F}_{A \rightarrow B} !$$

what happens if the dimensions of one of the

two bodies, say B, cannot be regarded as negligible?

such would be the case of a small body A and

the Earth B.

Then the body B can be

thought of as being

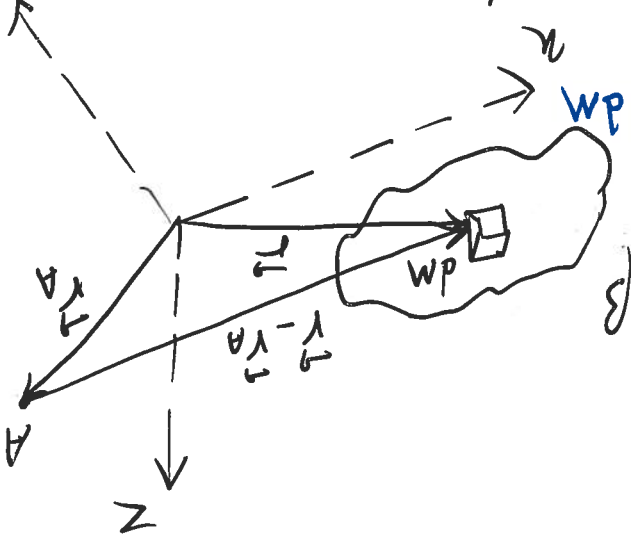
composed of a number

of small massive elements

of volume dV and

the attraction of each of these on A can be

Investigated separately. If the independent variable



is denoted by $\vec{r}$, the mass density within the body by $\sigma(\vec{r})$ and dm is chosen small enough so that σ in dm can be considered constant, then the following formula is obtained:

$$\vec{F}_{dm \rightarrow A} = G \frac{\sigma(\vec{r}) dm}{|\vec{r} - \vec{r}_A|^3} (\vec{r} - \vec{r}_A) \quad (4)$$

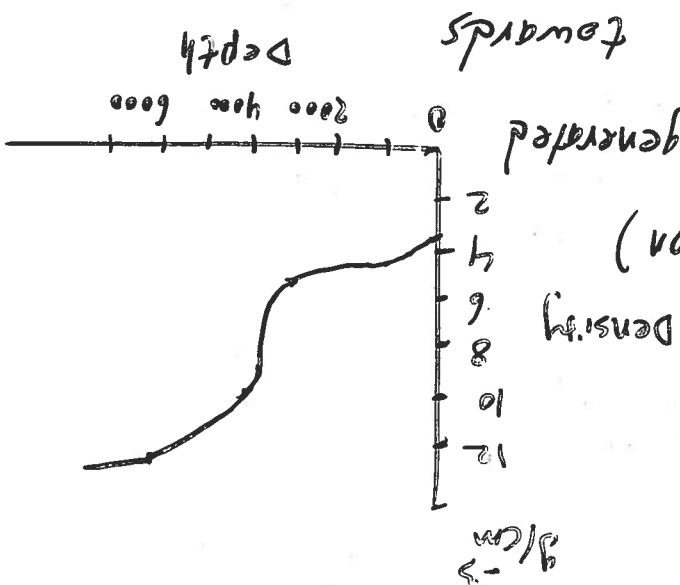
It has been determined by experiments that gravitational forces are additive. This means that the sum of forces produced by the elements dm is equal to the force exerted by the whole body B . By considering the volumes dm infinitesimally small, the final equation is obtained by integrating over the body B :

$$\vec{F}_{B \rightarrow A} = \vec{F}(\vec{r}_A) = Gm \iiint_B \frac{\sigma(\vec{r})}{|\vec{r} - \vec{r}_A|^3} (\vec{r} - \vec{r}_A) dV \quad (5)$$

This equation can be used to study the gravitational force of the Earth on bodies the dimensions of which can be considered negligible with respect to the Earth. To compute the gravitation, though, the density distribution $\sigma(\vec{r})$ with the earth must be known.

But such a distribution is known only approximately. (5)

It can be seen that the gravitational force produced by such an Earth model (perfectly spherical distribution) is radial, i.e., the force generated by this body always points towards the center of mass, and its magnitude depends only on the distance from the center of mass.



If the mean radius of the Earth (R) is taken equal to 6371.009 km, and GM equal to $3.986005 \times 10^{20} \text{ cm}^3 \text{ s}^{-2}$, the Equation (3) gives the mean value of the gravitational attraction at the surface of the Earth:

$$|\vec{F}| = F = 982.022 \text{ [cm s}^{-2}] \times m$$

where m is the mass of the attracted particle. Since the real distribution of density within the Earth is not only radially but also laterally irregular and the Earth is not spherical, the gravitational force field is not perfectly radial either. The value of gravitation above is only a mean global value.

In the absence of better knowledge of the actual density distribution, Equation (5) is of limited value to geodesy except to ^⑥ show, theoretically, how

gravitation depends on density. Equation (5) also shows that when the density varies with time, so does the gravitation force.

A convolution integral is the name for an integral over a function convolved with a kernel, i.e., of a product of the function with the kernel, while one argument of the kernel becomes the dummy variable, the other becomes the argument of the result of the integration.

$$f_3(y) = \int f_1(u) \cdot f_2(uy) du \quad (6)$$

$f_2(uy)$ is called the kernel (weight function) In Equation (5) $\frac{\vec{r} - \vec{r}_A}{|\vec{r} - \vec{r}_A|^3}$ is the kernel and

$\text{over } \vec{r}$ is the function which is convolved with ~~the kernel~~ ^{the kernel} ~~integral~~.

The fact that the Earth is spinning complicates things somewhat even if the Earth is assumed to be rigid. the spin of the Earth gives rise to an additional force. This force ($\vec{f}$), although only apparent in nature, can be observed to be acting on all earthbound objects ^⑦ that share the spin

with the Earth). It is called the centrifugal force. Its direction is always perpendicular to the

instantaneous spin axis and its nature is only

apparent because as soon as the object ceases to spin with the Earth — ceases to be earthbound — the centrifugal force vanishes. The magnitude f of the centrifugal force acting on a particle is

known to be equal to

$$f = p \omega^2 m$$

(7)

where p is the perpendicular distance of the particle from the spin axis, ω is the Earth's spin angular velocity, and m is the mass of the particle.

If the spin angular

$$\text{velocity } \omega = 72.92115 \times 10^{-6} \frac{\text{rad}}{\text{s}}$$

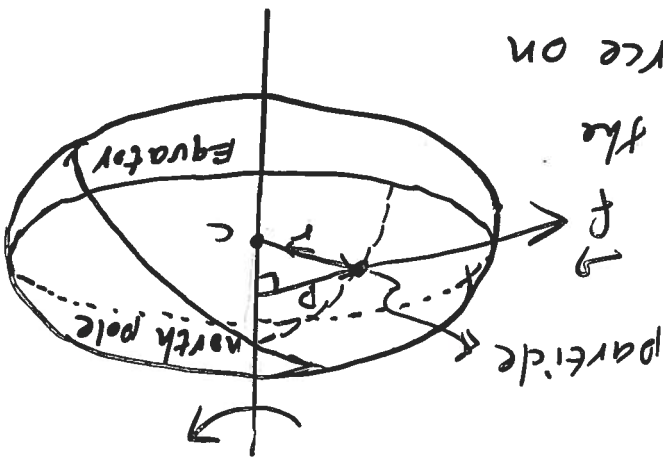
and $p = 6378.137 \text{ km}$, then the

value of the centrifugal force on

the equator is obtained

$$f \doteq 3.392 [\text{cm.s}^{-2}] \times m$$

which is about 0.35% of the gravitational force on the poles, the centrifugal⁽⁸⁾ force vanishes.

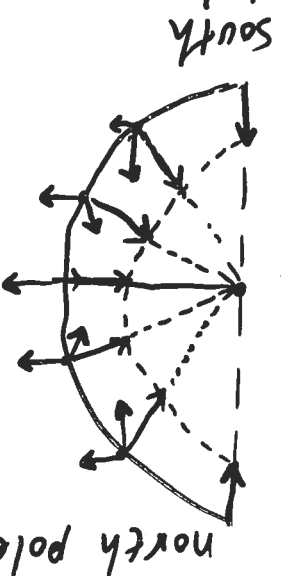


The sum of the gravitational and the centrifugal forces is called the gravity force.

It is easy to understand that the gravity force is stronger

on the poles than on the Equator. It is usual to work with

accelerations rather than forces.



$$\vec{F}'(\vec{r}_A) = \vec{F}_{\beta \rightarrow A} + \vec{F}_A = \left\{ G \iiint_{\beta} \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}_A|^3} (\vec{r} - \vec{r}_A) d\beta + \vec{P}_A \omega^2 \right\} m \quad (8)$$

pole

From Newton's second law ($\vec{F} = m\vec{a}$) $\Rightarrow$

$$\vec{F}'(\vec{r}_A) = \vec{g}(\vec{r}_A) m \quad (9)$$

where

$$\vec{g}(\vec{r}_A) = \left\{ G \iiint_{\beta} \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}_A|^3} (\vec{r} - \vec{r}_A) d\beta + \vec{P}_A \omega^2 \right\} \quad (10)$$

is called the gravity vector or gravity acceleration vector. In a study of the geometrical properties

of the gravity force field $\vec{F}'$, It is sufficient to concentrate on the acceleration $\vec{g}$. The mass m

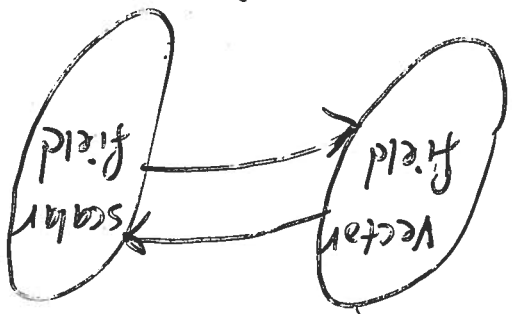
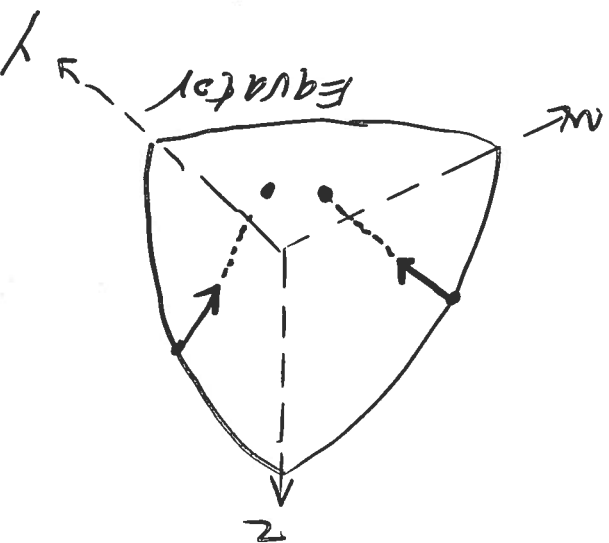
of the test particle can be regarded as the

scale of the field $\vec{F}'$, the gravity field $\vec{g}$ gives the complete geometrical picture of the gravity

force field. the gravity field, being a vector

field, possesses a magnitude (absolute value)

and a direction. A mean magnitude of gravity on the surface of the Earth is of the order of 980.3 Gal (one $\text{Gal} = \text{one cm/s}^2$).



The gravity field being a vector field means that there is a vector, i.e., a triple of numbers, assigned to every point in space. It is much more expedient to work with a scalar field, where there is just one number needed at every point. A question now arises: Is it possible to represent completely the vector field by a scalar field? The answer is yes, at least for some vector fields including the Earth's gravity field. Let us take an arbitrary, closed, spatial curve S within the vector field $\vec{v}$.

If the following equation

$$\oint_S \vec{v}(\vec{r}) \cdot d\vec{r} = 0$$

(11)

(10)

Irrotationality

Let us show that the gravitational field of the Earth is irrotational. It means that we have to show that

$$\oint_S \vec{F}(\vec{r}) d\vec{r} = 0$$

Another interpretation of above formula is that the work being done to overcome this force $\vec{F}$ is not dependent to the path.

$$W_{AB} = \int_B^A \vec{F}(\vec{r}) d\vec{r}$$

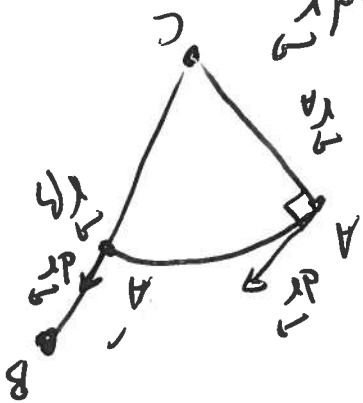
$$W_{AB} = - \int_B^A GM \frac{d\vec{r}}{r^3}$$

$$W_{AB} = - \int_B^A GM \frac{1}{r^3} d\vec{r} - \int_B^A GM \frac{1}{r^3} d\vec{r}$$

$$W_{AB} = -GM \int_B^A \frac{1}{r^3} d\vec{r} + GM \int_B^A \frac{1}{r^3} d\vec{r}$$

$$W_{AB} = 0 + GM \int_B^A \frac{dr}{r^2} = -GM \left[-\frac{1}{r} \right]_B^A$$

$$W_{AB} = -GM \left(-\frac{1}{r_B} + \frac{1}{r_A} \right) = GM \left(\frac{1}{r_B} - \frac{1}{r_A} \right)$$



This shows that the gravitational potential is not dependent to the path and is only dependent to the points A and B. then the gravitational field is irrotational

where $d\vec{r}$ is directed along the curve S , holds for any curve S , then the field $\vec{v}$ is called irrotational. If a field is irrotational, then there exists a scalar field V such that

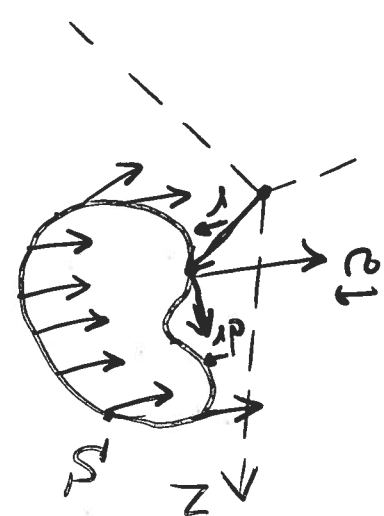
$$\vec{v}(\vec{r}) = \nabla V(\vec{r}) = \text{grad } V(\vec{r}) \quad (11)$$

If time is considered, then

for the above equation to hold, the

field $\vec{v}$ also has to be conservative, i.e., not

Y



changing with time. This scalar field V is called the potential energy of $\vec{v}$, and $\vec{v}$ is then the gradient of V . From the physical point of view, V is the amount of work needed to overcome the force $\vec{v}$. Its physical units are $\text{g cm}^2 \text{s}^{-2}$. It has been

shown that the gravity force field is irrotational, and as such, it has a corresponding potential.

Furthermore, because the gravity field $\vec{g}$ differs from the gravity force field only by a scale m , it is easy to see [from Eq. (11)] that the gravity field can be expressed as

$$\vec{F}' = m\vec{g} = \nabla V = m \nabla W$$

(12)

In other words, there also exists a scalar field W

(11)

such that

$$\vec{g} = \nabla W$$

(13)

This scalar field is known as the gravity potential. The gravity potential W is sometimes treated as the negative amount of work needed to overcome the gravity force ~~mg~~ acting on a unit mass m . But its physical units are $\text{cm}^2 \text{s}^{-2}$ and thus do not reflect the presence of any mass. It is preferable, therefore, to view W as work in the kinematic sense, i.e.

involving no mass. Since the potential W differs from the potential energy V only by scale m — the mass of the attracted particle — the geometries of the V and W scalar field are the same.

Let us have another look at Eq. (8): It can be seen that the gravity acceleration can be expressed as a sum of the triple integral, representing the gravitational acceleration, and another term

representing the centrifugal acceleration. Since the (gradient) differential operator ∇ is a linear operator, the gravity potential W can also be expressed as the sum of a gravitational potential

W_g and a centrifugal potential W_c . If the gravitational acceleration is denoted by $\vec{g}_g$ and the

(12)

centrifugal acceleration by $\vec{g}_c$, then

$$\vec{g} = \vec{g}_g + \vec{g}_c = \nabla w_g + \nabla w_c = \nabla (w_g + w_c) \quad (14)$$

considering the (gradient) differential operator ∇ as

$$\nabla = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \quad (15)$$

It can be verified by students, by evaluating the gradients and comparing with Eq. (8) that the gravitational potential w_g and centrifugal potential w_c are given by the following formula:

$$w_g(\vec{r}_A) = G \iiint_V \frac{\rho(\vec{r})}{|\vec{r} - \vec{r}_A|} dV \quad (16)$$

$$w_c(\vec{r}_A) = \frac{1}{2} \rho_A^2 \omega^2 \quad (17)$$

It is obvious that while w_g decreases above the Earth (i.e. for $|\vec{r}_A| > |\vec{r}|$), being inversely proportional to the distance $|\vec{r} - \vec{r}_A|$, w_c increases proportionally to the square of the distance ρ_A from the spin axis. For instance, taking a direction from the center of mass in the equatorial plane, it is found that the potential changes with distance, as shown in the Figure of next page. When interpreting this Figure, one has to bear in mind that the combined

potential w acts only on Earthbound bodies or particles. As soon as the body stops spinning with the Earth, the centrifugal potential w_c ceases to be relevant, and only the gravitational potential w_g prevails.

The gravity potential w must contain all the information there is about the gravity field.

Gravity force of Eq. (10) and gravity potential of

Eq. (13) is of limited value to geodesy. There are

some limitations for the determination of gravitational

parts of gravity force and potential (triple integral).

centrifugal force and potential determination is ok!

To evaluate this triple integral outside the Earth's

surface we have to have a very good information

about the density distribution $[\rho(r)]$. Also for

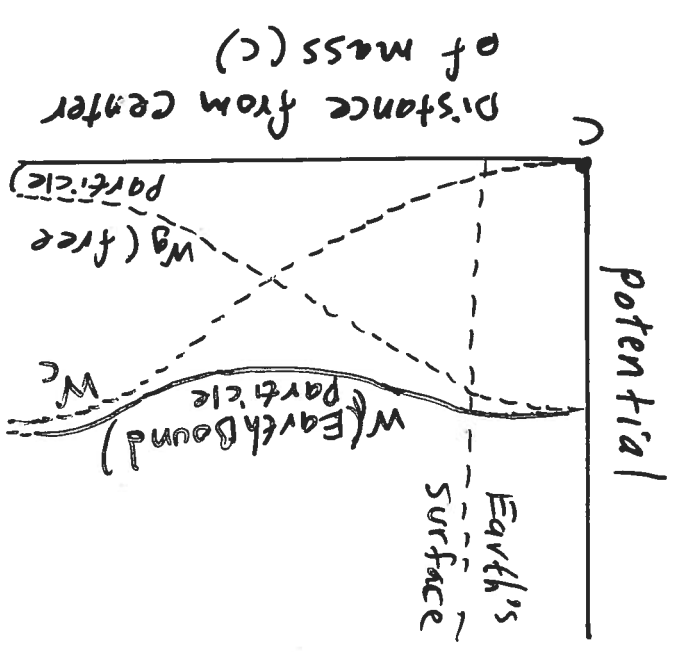
evaluating this triple integral we have to have a

good knowledge of mathematical figure of the Earth

(/3) - for a point inside the Earth a singularity

problem might happen where $r \rightarrow r_A$.

(14)



Hence we shall concentrate here on the ways of obtaining w . To derive the ^{another}

fundamental partial differential equation that describes the behaviour of the gravity potential w , let us first have a look at the local behaviour of the gravitational vector $\vec{g}$. Its behaviour at a point is fully described by its curl and divergence given in the neighbourhood of that point.

$$\text{div } \vec{a} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z} = \text{scalar} = \nabla \cdot \vec{a} \quad \text{scalar product}$$

$$\text{curl } \vec{a} = \left(\frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z} \right) \hat{x} + \left(\frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x} \right) \hat{y} + \left(\frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y} \right) \hat{z} = \text{vector} \quad \text{vector product}$$

Since the gravitational potential field is irrotational

and, in the first approximation, conservative, It means that we can immediately write:

$$\text{curl } \nabla w(\vec{r}_A) = \text{curl } \vec{g}(\vec{r}_A) = \nabla \times \vec{g}(\vec{r}_A) = 0 \quad (19)$$

for any point A , where $\vec{r}_A$ is a position vector in an arbitrary system of coordinates.

on the other hand, the divergence of $\vec{g}(\vec{r}_A)$ can be written as the limiting case of the Gauss formula:

$$\text{div } \vec{g}(\vec{r}_A) = \nabla \cdot \vec{g}(\vec{r}_A) = \lim_{V_A \rightarrow 0} \frac{\oint_{\partial V_A} \vec{g}(\vec{r}) \cdot \vec{n} \, ds}{V_A} \quad (20)$$

(15)

where V_A is the volume enclosed by the surface S , and $\vec{n}$ is a unit outward vector normal to S .

If the integral of the gravitational

flux $\vec{g} \cdot \vec{n} ds$ is positive, point A is called a source; if the integral is negative, then A is called a sink. At the outset, this is the important question: Is the divergence of the gravitational acceleration positive or negative?

To work out the answer, let us first denote the mass M of volume V_A embraced by the surface S as

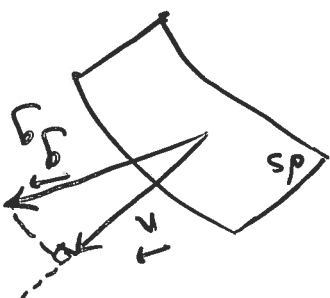
$$M = V_A \rho(\vec{r}_A) \quad (21)$$

with the understanding that for a sufficiently volume V_A the mass density ρ is uniform (constant) in V_A . If for simplicity, the origin of the coordinate system is put at the center of V_A , we discover that V_A generates an acceleration $\vec{g}_g$ given by:

$$\vec{g}_g(\vec{r}_A) = \frac{GM}{r_A^3} \vec{r}_A \quad (22)$$

It can further be shown that when the limits is

considered in Eq. (20), the shape of the surface S is immaterial as long as it fully encloses the volume V_A . Thus for simplicity a spherical surface



can be chosen. In this case it is clear that the $\vec{g}_g$ generated by the spherical volume V_A is normal to S everywhere, and $\vec{g}_g \cdot \vec{n}$ is equal to g_g . The gravitational flux can then be written as:

$$\vec{g}_g \cdot \vec{n} ds = g_g ds = -\frac{GM}{r^2} ds = -\frac{G \varphi(r_A) V_A}{r^2} ds \quad (23)$$

where r is the radius of the sphere S . With the realization that the surface of a sphere of radius r is equal to $4\pi r^2$, it is not difficult to see that the integral of the gravitation flux gives:

$$\oint_S \vec{g}_g \cdot \vec{n} ds = -4\pi G \varphi(r_A) V_A \quad (24)$$

and that upon dividing by V_A and taking the limit we obtain:

$$\text{div } \vec{g}_g(r_A) = -4\pi G \varphi(r_A) \quad (25)$$

This equation which can also be obtained directly from the definition of divergence by taking the limiting case of $r \rightarrow 0$, is valid, of course, in any coordinate system. Its interpretation is as follows:

since $\varphi(r_A)$ is a non-negative quantity, A can be either a sink of gravitational acceleration $\vec{g}_g(r_A)$ [if $\varphi(r_A) > 0$] or else the divergence of $\vec{g}_g(r_A)$ is zero in

A [if $\vec{a}(\vec{r}_A) = 0$]. Having disposed of the local

behaviour of gravitational acceleration, let us now turn to centrifugal acceleration $\vec{g}_c(\vec{r}_A)$.

The centrifugal force is irrotational, and

$$(26) \quad \text{curl } \vec{g}_c(\vec{r}_A) = \vec{0}$$

The divergence of $\vec{g}_c(\vec{r}_A)$ is easily evaluated by

selecting the coordinate system (x y z) so that

the z-axis coincides with the spin axis of the

Earth. Recalling (7) and realizing that $p = \sqrt{x^2 + y^2} = a \cos \theta$

we get

$$(27) \quad \vec{g}_c(\vec{r}_A) = \omega^2 (x\vec{i} + y\vec{j})$$

and the divergence is readily obtained as

$$(28) \quad \text{div } \vec{g}_c(\vec{r}_A) = 2\omega^2$$

It can be seen that the divergence of

centrifugal acceleration remains constant ($2\omega^2$)

throughout the space and does not depend on the

coordinate system.

Realizing that gravity acceleration $\vec{g}_g(\vec{r}_A)$ is the

sum of the two accelerations treated above, we obtain

as a consequence of the linearity of the ∇ operator

$$(29) \quad \nabla \cdot \vec{g}(\vec{r}_A) = \text{div } \vec{g}_g(\vec{r}_A) = -4\pi G \rho(\vec{r}_A) + 2\omega^2$$

Further, since $\vec{g} = \nabla W$, we finally get

(18)

$$\Delta^2 W(\vec{r}_A) = -4\pi G \rho(\vec{r}_A) + 2w^2$$

(30)

$$\Delta^2 = \Delta^2!$$

(31)

$$\text{where } \Delta^2 a = \Delta \cdot (\Delta a) = \frac{\partial^2 a}{\partial x^2} + \frac{\partial^2 a}{\partial y^2} + \frac{\partial^2 a}{\partial z^2}$$

This is the fundamental, partial differential equation (of second order) for gravity potential that we set out to derive. It is recognized as being the Poisson Equation.

Although Eq. (30) is valid in this form throughout the space, it is sometimes expedient to spell out its

two special forms:

a) one applies in empty space ($\rho=0$) and reads:

$$\Delta^2 W(\vec{r}) = 2w^2 \quad (32)$$

Disregarding the density of the atmosphere, which is negligibly small compared with the density of the

Earth, we can say that Eq. (32) is valid outside the Earth.

b) The second special case of (30) can be formulated

for the surface of the Earth. A point A located on the surface of the Earth may be considered as having a density equal to a fraction of the density at a point

which is entirely within the Earth. In other words, any differential neighbourhood of point A on the

Earth's surface is partially empty and partially

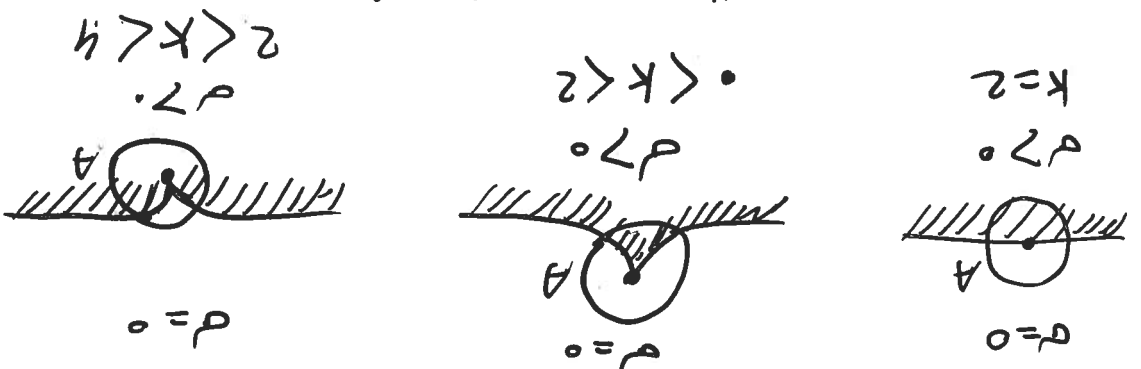
Hence.

$$\nabla^2 w(r_A) = -k\pi G \rho(r_A) + 2w^2$$

(33)

where $\rho(r_A)$ is understood to be the density of

the material at the Earth's surface and $k \in (0, 4)$



These equations illustrate the fact that the second derivatives of gravity potential (contained in the

left-hand sides) are discontinuous on a boundary between any two media of different densities,

the Earth's surface being a notable example. It may also be noted that the fundamental differential

equation for gravitational potential w_g outside the Earth's surface (in empty space), known as the potential equation and also is called as Laplace

Equation:

$$\nabla^2 w_g = \Delta w_g = 0$$

(34)

A solution of Laplace's Equation is known as a harmonic function; It has many useful properties. The gravitational potential in empty space is a harmonic function!

(20)

clearly, the problem of determining the Earth's gravity field can be broken down to the determination of the gravitational field and the determination of the centrifugal field. while the former task is indeed formidable, the determination of the centrifugal field is not: g_c is a simple function of position since the Earth-spin angular velocity is known very accurately. consequently, in the reminder of this course, we shall concentrate on the gravitational field.

Geodetic interest lies mainly in the space immediately outside the Earth (to an altitude of a few thousand kilometers), the Earth's surface, and the uppermost layer of the Earth's crust. From the

character of the differential equations valid within and outside the Earth, one can see that it is expedient to split the region of interest into two: outside and inside the Earth. Accordingly, we outside the Earth is often spoken of as the Earth's external gravitational potential and inside the

Earth as the internal gravitational potential. one of the most important properties of harmonic function is related to Dirichlet's boundary value

(21)

problem.

Dirichlet's principle says that if a harmonic function f outside a known boundary is unknown but it has a known values on the boundary, a unique solution for the f outside the boundary can be derived. It has been shown to have a unique solution if the boundary should be smooth enough. Therefore, at first glance, the determination of the Earth's gravitational potential w_g at least outside the Earth, should not be very difficult. External w_g satisfies the Laplace Equation outside the Earth, and it should thus be enough to obtain the values of the gravitational potential w_g on the surface of the Earth (Earth = known boundary; values of w_g on the Earth as the known values on boundary $\Rightarrow$ Dirichlet's Boundary value problem).

and solve for w_g outside the Earth using a technique for solving the Laplace boundary value problem.

Unfortunately, the Earth's surface is not sufficiently smooth to ensure the uniqueness of such an external solution. But there are possible ways of overcoming this obstacle. Therefore, for the present,

we acknowledge that there is possible to obtain a unique solution and turn our attention to the available techniques for solving the boundary value problem. Before that, to complete the BVP we spell out three boundary value problem

1-Boundary value problem of the first kind =

Dirichlet's boundary value problem = f unknown outside a known boundary, but the values of f

on the boundary are known $\Rightarrow$ find f outside the boundary

2-Boundary value problem of the second kind =

Neumann's boundary value problem = f unknown

outside a known boundary, but the values of f the

normal derivative of f (the derivative with respect to the normal to boundary) are known $\Rightarrow$ find f

outside the boundary

3-Boundary value problem of the third kind =

mixed type of boundary value problem = f unknown

outside a known boundary, but a combination of

the values of f and the values of the normal

derivative of f on the boundary is known $\Rightarrow$

find f outside the boundary

(23)

Dirichlet's principle :

Let us have an unknown function of f_1 which is harmonic in region A and it has boundary values of f_1 which are known. Also, we have an unknown function of f_2 which is harmonic in the same region A , and it has the same boundary values as f_1 , which

are known.

$$\begin{aligned}\Delta^2 f_1 &= \frac{\partial^2 f_1}{\partial x^2} + \frac{\partial^2 f_1}{\partial y^2} + \frac{\partial^2 f_1}{\partial z^2} = 0 \\ \Delta^2 f_2 &= \frac{\partial^2 f_2}{\partial x^2} + \frac{\partial^2 f_2}{\partial y^2} + \frac{\partial^2 f_2}{\partial z^2} = 0\end{aligned}$$

$$\Delta^2 f_1 - \Delta^2 f_2 = \Delta^2 (f_1 - f_2) = \Delta^2 f = 0$$

we know that the boundary values of f is equal to zero. According to a property of harmonic function: the maximum and

minimum values of a harmonic function

always occur on the boundary. Then function of f outside the boundary is also equal to

$$\text{zero} \Rightarrow f_1 - f_2 = f = 0 \Rightarrow f_1 = f_2 \text{ then}$$

Dirichlet's principle (22a) ensures a unique solution.

From among many existing techniques to solve the

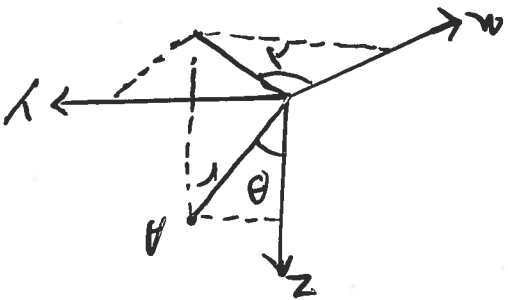
following boundary value problem

$$\nabla^2 w_g(r, \theta) = -k \pi G \sigma \quad ; k = \begin{cases} 0 & \text{A outside boundary} \\ 2 & \text{A on smooth boundary} \\ 4 & \text{A inside boundary} \end{cases}$$

(35)

the two that are mostly used: transformation to integral Equations, and the method of separation of variables (the Fourier technique). The Fourier technique is used in this course. To obtain the solution in its simplest form, it is expedient to select a more convenient coordinate system than the Cartesian. Two such systems are commonly being used - the spherical and the ellipsoidal.

The spherical coordinate system are used in this course. (r, θ, λ) are spherical coordinates. The spherical coordinates are very simple and convenient to solve the



Laplace Equation of $\nabla^2 w_g = 0$.

The Laplace Equation in spherical coordinate system

can be written as.

$$\nabla^2 w_g(r, \theta) = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial w_g}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 w_g}{\partial \lambda^2} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial w_g}{\partial \theta} \right) = 0$$

(24)

(36)

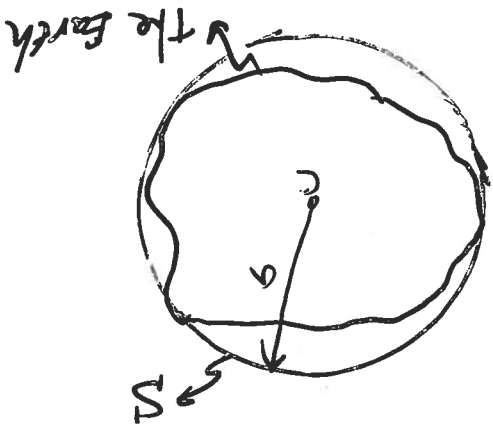
Now we are ready to apply the method of separation of variables to the Eq. (36). According to the idea of this method we rewrite the gravitational potential $w_g(r, \theta, \lambda)$ in terms of a product of the following three functions:

$$w_g(r, \theta, \lambda) = R(r) \cdot T(\theta) \cdot L(\lambda) = R(r) \cdot J(\theta, \lambda)$$

If the spherical coordinate system (r, θ, λ) is

selected so that the origin is close to the center of mass of the Earth, the surface of the Earth will be very close to the sphere $r=R$. Then the first function R , will describe the behaviour of w_g in the direction approximately normal to the Earth's surface (globally speaking), while the function J will depict the variations of the potential on the Earth's surface. More precisely, if we take a sphere S of radius $r=a$, called the Brillouin sphere, that encompasses all the masses of the Earth, then on this sphere R will be constant and all the variation of w_g on S will be characterized by J .

(25)



clearly, disregarding the atmosphere, Eq. (36)

is valid outside this sphere.

The first application of the Fourier method to

Eq. (36) yields :

$$(38) \quad r^2 R'' + 2rR' - c_1 R = 0$$

and

$$(39) \quad \cot \theta \frac{\partial^2 \theta}{\partial \theta^2} + \frac{\partial^2 \theta}{\partial \lambda^2} + \frac{1}{\sin^2 \theta} \frac{\partial^2 \theta}{\partial \lambda^2} + c_1 \theta = 0$$

Equation (38) is an ordinary differential Equation of second order; the primes denote derivatives

with respect to r . Equation (39) is still a partial differential Equation in θ and λ . The

second application of the Fourier method to

Eq. (39) yields :

$$(40) \quad \sin^2 \theta T'' + \sin \theta \cos \theta T' + (c_1 \sin^2 \theta - c_2) T = 0$$

and

$$(41) \quad L'' + c_2 L = 0$$

In these two new equations, the variables θ

and λ are separated. The system of the three

ordinary differential equation (38), (40) and (41)

is equivalent to Eq. (36). their interrelation is

(26)

ensured through the two common constants c_1 and c_2 . the three ordinary differential equations give the general solution for the functions R, T and L .

Equation (41) is readily recognized as an Equation of simple harmonic motion. Its solution for $c_2 = m^2$

for $m = 0, 1, 2, 3, \dots$ can be written as: $L = \cos m\lambda$; $L = \sin m\lambda$
 eigenfunction $\downarrow$ eigenvalues $\downarrow$
 (42)

Any linear combination L of these eigenfunctions satisfies (41).

To solve Eq. (40), we substitute t for $\cos\theta$ and

m^2 for c_2 in Eq. (40):

$$(1-t^2)T'' - 2tT' + \left(c_1 - \frac{m^2}{1-t^2}\right)T = 0 \quad (43)$$

where the derivatives are now taken with respect

to t . Eq. (43) is a second-order Legendre equation

for T . It makes sense to attempt its solution

only for such values of m for which we know the

solution of (41) exists. It has been shown that $\downarrow$

Eq. (43) gives a solution only for the following c_1

$$c_1 = n(n+1) ; n = m, m+1, m+2, \dots$$

when $\theta \in (0, \pi)$.
 (27)

These are the admissible eigenvalues of the Legendre equation. corresponding solutions for eigenfunction

the Legendre equation are the Legendre associated functions (of n th degree and m th order)

$$(44) \quad P_m(t) = (1-t^2)^{m/2} \frac{d^m}{dt^m} P_n(t)$$

where

$$(45) \quad P_n(t) = \frac{1}{2^n n!} \frac{d^n}{dt^n} (t^2-1)^n$$

are the Legendre functions. It should be noted that the solutions are orthogonal for $t \in (-1, 1)$ [or $\theta \in (0, \pi)$] with weight $w(t) = 1$.

Now we can see that Eq. (39) is satisfied by

any function J that is a product of a linear

combination of trigonometric functions, with a

linear combination of Legendre associated functions (44) for the admissible values of m and n . Such a

product can generally be written as :

$$J(\theta, \lambda) = \sum_{n=0}^{\infty} \sum_{m=0}^n (A_{nm} \cos m\lambda + B_{nm} \sin m\lambda) P_m(\cos \theta) = \sum_{n=0}^{\infty} \sum_{m=0}^n (A_{nm} Y_{nm} + B_{nm} Y_{nm}^s) \quad (46)$$

where A_{lm} , B_{lm} are some arbitrary numbers. the functions $Y_{lm}^c = \cos m \lambda f_{lm}(\cos \theta)$ and $Y_{lm}^s = \sin m \lambda f_{lm}(\cos \theta)$

are called (surface) spherical harmonic functions. They can be regarded as solutions (eigenfunctions) of the Laplace equation in spherical coordinates on the surface of a sphere. the integers

$$\begin{cases} m=0,1,2,\dots \\ n=m, m+1, m+2 \end{cases} \text{ or identically } \begin{cases} m=0,1,2,\dots \\ n=0,1,2,\dots \end{cases}$$

are then the eigenvalues of the Laplace equation. Note that the spherical harmonics, being eigenfunctions, are indeed orthogonal on a spherical surface S , i.e. for any $(\theta, \lambda) \in (0, \pi) \times (-\pi, \pi)$. this means that the following Equation

$$\oint_S Y_{lm}^c(\theta, \lambda) Y_{l'm'}^c(\theta, \lambda) d\Omega = \delta_{ll'} \delta_{mm'} \quad (47)$$

where $d\Omega$ is the solid angle element, is satisfied for $l' \neq l$ or $m' \neq m$ or $k \neq l$.

$$\iint_S = \int_0^{2\pi} \int_0^\pi \quad \text{for the integral over the sphere.}$$

the expression $d\Omega = \sin \theta d\theta d\lambda$ denotes the surface element of the unit sphere or the element of

solid angle, a solid angle being defined as the corresponding area on the unit sphere. (29)

It can be shown that the norm of surface spherical harmonic functions are equal to

$$\|Y_{nm}\|^2 = \oint_S Y_{nm}^2 dv = \begin{cases} \frac{4\pi}{2n+1} & m=0 \\ \frac{2\pi}{2n+1} \frac{(n+m)!}{(n-m)!} & m \neq 0 \end{cases} \quad (48)$$

for both superscripts c and s . Functions

$$\overline{Y_{nm}} = \frac{Y_{nm}}{\|Y_{nm}\|} \text{ are then orthonormal, i.e.,}$$

$\|\overline{Y_{nm}}\| = 1$, and are thus called (fully) normalized

spherical harmonic functions. Spherical harmonic functions for $m=0$ are called zonal harmonics

for $n=m \neq 0$ are called sectorial harmonics and

for $n \neq m \neq 0$ are called tesseral harmonics.

Returning to the main problem at hand, the exterior problem for w_g (outside the Brillouin sphere), we

can now use spherical harmonics to solve it

quite easily. Consider Eqs. (37) and (46), the

solution clearly is as

$$w_g(r, \theta, \lambda) = R(r) \sum_n \sum_{m=-n}^n (A_{nm} Y_{nm}^c(\theta, \lambda) + B_{nm} Y_{nm}^s(\theta, \lambda)) \quad (49)$$

here, the coefficients A_{nm} and B_{nm} have to be

determined in such a way as to satisfy the

boundary value $w_g(r=a, \theta, \lambda)$ on the sphere.
that means the following equation would have to
be satisfied:

$$(50) \quad w_g(a, \theta, \lambda) = R(a) \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} (A_{nm} Y_{nm}(\theta, \lambda) + B_{nm} Y_{nm}(\theta, \lambda))$$

where, of course, $R(a)$ is constant. We are now
ready to focus our attention on the third ordinary
differential equation (38). Here, again, it is

meaningful to seek the solution only for the

admissible values of $c = n(n+1)$. It is [Eq. (38)]

a variety of an Euler equation and is readily

solved using substitution of $\exp(t)$ for r . We

find that two families of functions, R_1 and R_2

$$(51) \quad R_1(r) = \exp(nt) = r^n; \quad R_2(r) = \exp((-n-1)t) = r^{-(n+1)}$$

satisfy the Equation (38) for any admissible n .

The first functions increase beyond limits when r

grows beyond limits. This is inconsistent with the

expected behaviour of the gravitational potential. Hence

when a solution to the external problem is sought,

we are obliged to take the second family of

functions ($R_2(r)$) to make the solution compatible with

physical requirements. For parallel reasons, when the

Laplace interior problem (a function is harmonic

inside a certain sphere) is contemplated, R_1 is used. It is expedient to make the radial function $R=R_2$ unitless so that only the coefficients A_{nm} and B_{nm} in (49) are the carriers of physical unit. This is normally done by scaling the spherical coordinate system so as to make the radius of the boundary sphere a unity: instead of using r , we use $\frac{r}{a}$. In this coordinate system, the radial function becomes:

$$R_2 = r^{-(n+1)} \Rightarrow \text{if } r \rightarrow \frac{r}{a} \Rightarrow R_2 = \left(\frac{r}{a}\right)^{-(n+1)} = \left(\frac{r}{a}\right)^{n+1}$$

$$n = 0, 1, 2, \dots$$

such a scaling, of course, affects even the values

of the coefficients in (49) and thus preserves the correct overall magnitude of w_g . Therefore

the final form for the external gravitational

$$\text{potential is: } w_g(r, \theta, \lambda) = \sum_{n=0}^{\infty} \sum_{m=0}^n \left(\frac{r}{a}\right)^{n+1} [A_{nm} \cos m\lambda + B_{nm} \sin m\lambda] P_m(\cos \theta)$$

This equation is equivalent to integral equation (16), the only difference being that instead of the

unknown mass density σ , here we have infinitely

many unknown coefficients A_{nm} and B_{nm} (in the units of potential). Equation (53) is known as the development of gravitational potential into spherical harmonics, or development into the eigen functions of the Laplace equation in spherical coordinates.

If the earth's gravitational potential was a function harmonic inside the Brillouin sphere we had the

spherical harmonic expansion as

$$w_g(r, \theta, \lambda) = \sum_{n=0}^{\infty} \sum_{m=0}^n \left(\frac{r}{a}\right)^n [A_{nm} \cos m\lambda + B_{nm} \sin m\lambda] P_{nm}(\cos \theta) \quad (54)$$

Before getting into the realm of determination of

the unknown coefficients, let us reflect on the role of the radial terms $(\frac{r}{a})^{n+1}$ outside the boundary sphere, their magnitudes diminish with increasing order n , since $\frac{r}{a} < 1$. This means that with increasing

altitude, the higher frequency (n and m) wavelength features of the potential field tend to be smoothed

out.

The question that may linger in the mind is what has been gained by deriving Eq. (53) instead of (16),

since we have succeeded only in replacing one unknown function of location (σ) by a whole series of unknowns (the coefficients). In fact, we have progressed at least one step further, because we are now able to evaluate the coefficients from data on and above the Earth instead of being dependent on the knowledge of mass distribution inside the Earth. If the value of the gravitational potential on the boundary sphere $S(r=a)$ is known, i.e., if

$$V(a, \lambda) = W_g(a, \theta, \lambda)$$

(55)

is known, then the potential coefficients A_{nm} and B_{nm} could be obtained through the standard procedure of developing V into spherical harmonics.

$$V(\theta, \lambda) = \sum_{n=0}^{\infty} \sum_{m=0}^n \overline{A_{nm}} \overline{Y_{nm}^c} + \sum_{n=0}^{\infty} \sum_{m=0}^n \overline{B_{nm}} \overline{Y_{nm}^c}$$

$$\Rightarrow \oint_S V(\theta, \lambda) \overline{Y_{kr}^c} ds = \oint_S \sum_{n=0}^{\infty} \sum_{m=0}^n \overline{A_{nm}} \overline{Y_{nm}^c} + \sum_{n=0}^{\infty} \sum_{m=0}^n \overline{B_{nm}} \overline{Y_{nm}^c} \oint_S \overline{Y_{kr}^c} ds$$

$$= \sum_{n=0}^{\infty} \sum_{m=0}^n \overline{A_{nm}} \oint_S \overline{Y_{nm}^c} \overline{Y_{kr}^c} ds + \sum_{n=0}^{\infty} \sum_{m=0}^n \overline{B_{nm}} \oint_S \overline{Y_{nm}^c} \overline{Y_{kr}^c} ds$$

$$= \overline{A_{kr}} \cdot 4\pi$$

$$\Rightarrow \overline{A_{kr}} = \frac{1}{4\pi} \oint_S V(\theta, \lambda) \overline{Y_{kr}^c} ds$$

$$\overline{B_{kr}} = \frac{1}{4\pi} \oint_S V(\theta, \lambda) \overline{Y_{kr}^c} ds$$

(54)

(56)

orthogonality and normalization of Legendre's polynomials and spherical harmonics

for $0 \leq n < \infty$, $0 \leq m \leq n$, it can be proved that

$$\frac{1}{2} \int_{-1}^1 P_n(t) P_m(t) dt = \begin{cases} \frac{1}{2n+1} \frac{(n+m)!}{(n-m)!} & \text{if } n=m \\ 0 & \text{if } n \neq m \text{ or } m \neq n \end{cases}$$

which indicates that $P_n(t)$ forms an orthogonal but not normalized system of polynomials. The normalized Legendre's polynomials are then defined

as:

$$\bar{P}_n(t) = f_n P_n(t)$$

where f_n is a factor defined as

$$f_n = \begin{cases} \sqrt{2n+1} & \text{if } m=0 \\ \sqrt{2(2n+1)} \frac{(n-m)!}{(n+m)!} & \text{if } m \neq 0 \end{cases}$$

Now we have:

$$\frac{1}{2} \int_{-1}^1 \bar{P}_n(t) \bar{P}_m(t) dt = \begin{cases} 1 & \text{if } n=m \\ 0 & \text{if } n \neq m \text{ or } m \neq n \end{cases}$$

$$2 \text{ if } n=m \text{ and } m \neq 0$$

The last result 2 for $n=m$ and $m \neq 0$ implies

34a

that $\bar{p}_m(t)$ is still not a completely normalized orthogonal system. But this "partial" normalization

is aimed at a complete normalization of the

surface spherical harmonics $Y_m(\theta, \lambda)$, for which

we have:

$$\frac{1}{4\pi} \iint_{\sigma} Y_{n'm'}(\theta, \lambda) Y_{nm}(\theta, \lambda) d\sigma = \begin{cases} 1/2 & \text{if } n=n' \text{ and } m=m' \\ 0 & \text{if } n \neq n' \text{ or } m \neq m' \end{cases}$$

Therefore, the normalized surface harmonics is

defined as:

$$\bar{Y}_{nm}(\theta, \lambda) = \sqrt{2} Y_{nm}(\theta, \lambda)$$

which satisfies:

$$\frac{1}{4\pi} \iint_{\sigma} \bar{Y}_{n'm'}(\theta, \lambda) \bar{Y}_{nm}(\theta, \lambda) d\sigma = \begin{cases} 1 & \text{if } n=n' \text{ and } m=m' \\ 0 & \text{if } n \neq n' \text{ or } m \neq m' \end{cases}$$

using the normalized surface spherical harmonics

$\bar{Y}_{nm}(\theta, \lambda)$, the gravitational potential $W_g(r, \theta, \lambda)$

can be written as:

$$W_g(r, \theta, \lambda) = \frac{GM}{r} \left[1 - \sum_{n=1}^{\infty} \left(\frac{a}{r} \right)^n \left(\sum_{m=0}^n (\bar{J}_m \cos m\lambda + \bar{K}_m \sin m\lambda) \right) \bar{Y}_{nm}(\cos \theta) \right]$$

(34b)

where $\overline{J_{nm}}$ and $\overline{k_{nm}}$ denotes the normalized spherical harmonic coefficients which are related to the un-normalized coefficient J_{nm} , k_{nm}

by

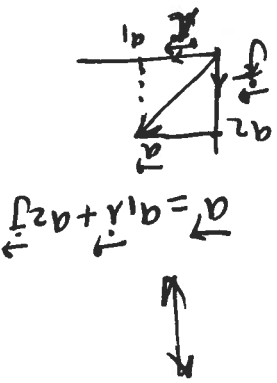
$$\overline{J_{nm}} = \frac{J_{nm}}{f_{nm}} \quad ; \quad \overline{k_{nm}} = \frac{k_{nm}}{f_{nm}}$$

$$\overline{J_{nm}} = -\frac{GM}{a_{nm}} \quad \overline{k_{nm}} = -\frac{GM}{a_{nm}}$$

may be it should be the right place to show that the gravitational potential $w_g(r, \theta, \lambda)$ can be shown in other forms too:

$$w_g(r, \theta, \lambda) = \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^{n+1} \sum_{m=0}^n (A_{nm}(\cos \theta) + B_{nm} \sin m\lambda) P_n(\cos \theta)$$

$$= \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^{n+1} \sum_{m=-n}^n D_{nm} Y_{nm}(\theta, \lambda)$$



with

$$D_{nm} = \begin{cases} A_{nm} & \text{for } m \leq 0 \\ B_{nm} & \text{for } m > 0 \end{cases}$$

$$Y_{nm}(\theta, \lambda) = \begin{cases} \cos m\lambda P_n|m|(\cos \theta) & \text{for } m \leq 0 \\ \sin m\lambda P_n m(\cos \theta) & \text{for } m > 0 \end{cases}$$

$\left(\frac{a}{r}\right)^{n+1} Y_{nm}(\theta, \lambda)$ is often called the solid spherical

harmonics.

(345)

As it is obvious for the spherical harmonic computations of Earth's gravitational field, we need to

compute the Legendre polynomials $P_m(t)$ and normalized Legendre's polynomials. Two formulas for computations of $P_m(t)$ and $P_m(t)$ for different n, m which are very simple for programming are given below:

$$P_n(t) = -\frac{n-1}{n} P_{n-2}(t) + \frac{n}{2n-1} t P_{n-1}(t)$$

$$P_m(t) = -\frac{n+m-1}{n-m} P_{n-2,m}(t) + \frac{n-m}{2n-1} P_{n-1,m}(t)$$

$$n \geq 1; m = n-1$$

$$P_{n,n-1}(t) = (2n-1)t P_{n-1,n-1}(t)$$

$$n \geq 2; m = n$$

$$P_{nn}(t) = (2n-1)\sqrt{1-t^2} P_{n-1,n-1}(t)$$

and for normalized one:

$$P_n(t) = -\frac{n}{\sqrt{2n+1}} P_{n-1}(t) + \frac{n}{\sqrt{2n+1}} t P_{n-1}(t)$$

$$P_{nm}(t) = -\sqrt{\frac{(2n+1)(n+m-1)(n-m-1)}{(2n-3)(n+m)(n-m)}} P_{n-2,m}(t) + \sqrt{\frac{(n+m)(n-m)}{(2n+1)(2n-1)}} t$$

$$P_{n-1,m}(t) \times P_{n-1,m}(t)$$

$$n \geq 1; m = n-1$$

$$P_{n,n-1}(t) = \sqrt{2n+1} t P_{n-1,n-1}(t)$$

$$P_{nn}(t) = \sqrt{\frac{2n+1}{2n}} \sqrt{1-t^2} P_{n-1,n-1}(t)$$

3rd

$$n \geq 2; m = n$$

These are recursive formulas and need some initial values:

n	m	1	2	3
$p_m(t)$	0	0	1	$\sqrt{1-t^2}$
$p_m(t)$	1	0	0	t
$p_m(t)$	2	0	1	$\sqrt{3}\sqrt{1-t^2}$
$p_m(t)$	3	0	0	$\sqrt{5}\left(\frac{3}{2}t^2 - \frac{1}{2}\right)$
$p_m(t)$	4	1	0	$\sqrt{5}t\sqrt{1-t^2}$

The GRS80 coordinate system which is defined based on the WGS84 ellipsoid has the following parameter values:

$$a = 6378137.000 \text{ m}$$

$$b = 6356752.3141 \text{ m}$$

$$f' = 298.257222101$$

$$e^2 = 0.006694380033$$

$$R = 6371000.7900 \text{ m}$$

$$GM = 3.986005 \times 10^{14} \text{ cm}^2 \text{ s}^{-2}$$

$$J_2 = -0.00108662$$

(34e)

Some notes in spherical harmonic expansion

consider the earth's gravitational potential:

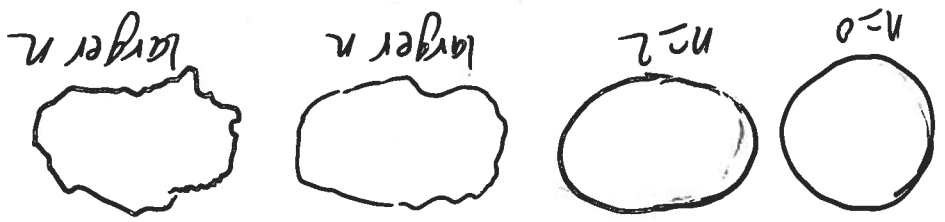
$$W_g(r, \theta, \lambda) = \sum_{n=0}^{\infty} \sum_{m=0}^n \left(\frac{a}{r}\right)^{n+1} [A_{nm} \cos m\lambda + B_{nm} \sin m\lambda] P_{nm}(\cos \theta)$$

The degree order n , also called wave number has an important role.

small n = long wavelength = short frequency = global information
 large n = short wavelength = long frequency = detail/(local) information

It means that small values of n show the global and general effect of gravitational potential and large

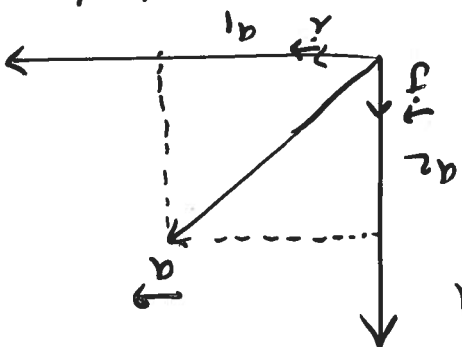
values of n show the local and detailed information of gravitational potential. For example $n=0$ shows the effect of a spherical Earth and $n=2$ shows the effect of an Ellipsoidal Earth and larger n shows the detailed information of the Earth.



34f

other thing which should be mentioned here is the rule of $\underline{Y}_m^c$ and $\underline{Y}_m^s$ in spherical harmonic presentation of gravity field parameters.

let us, for sake of comparison have a look at presentation of a vector in cartesian Euclidian



space. vector $\underline{a}$ can be written in the following form:

$$\underline{a} = a_1 \underline{i} + a_2 \underline{j}$$

where $\underline{i}$ and $\underline{j}$ are normalized base vectors which are

used to expand the vector $\underline{a}$ and a_1 and a_2 are the components of the $\underline{a}$ on the base vector directions.

The situation is the same for the spherical harmonic

representation of e.g. gravitational potential:

$$W_g(\underline{r}) = \frac{GM}{r} \left[1 + \sum_{n=1}^{\infty} \left(\frac{a}{r} \right)^n \left(\sum_{m=-n}^n \underline{Y}_m^c(\underline{r}) \underline{Y}_m^c + \underline{Y}_m^s(\underline{r}) \underline{Y}_m^s \right) \right]$$

This means that the fully normalized surface spherical harmonics $\underline{Y}_m^c$ and $\underline{Y}_m^s$ have the same role as $\underline{i}$ and $\underline{j}$ and are the base functions, but on the sphere and

$\underline{a}_m$ and $\underline{k}_m$ are the components of the $W_g(\underline{r})$ on the sphere

(349)

these are the equations for potential coefficients given here for simplicity when normalized spherical harmonic functions are used.

since there is an implied relationship between the density σ and the potential coefficients Ann and Bnn, one is tempted to have a

closer look at this relation. For instance, can knowledge of the potential coefficients teach us something about the internal distribution of masses? To answer this question, let us first discuss the (direct) relation of the coefficients with density. Equation (16) can be written

as:

$$W_g(\vec{r}_A) = G \iiint \frac{\sigma(\vec{r})}{r} dV \quad (57)$$

By denoting the spatial angle between $\vec{r}_A$ and $\vec{r}$ by ψ , the inverse of the distance between the point of interest A and the dummy point V can be expressed as:

$$P^{-1}(\vec{r}_A, \vec{r}) = (r_A^2 + r^2 - 2r r_A \cos \psi)^{-1/2} = \frac{1}{r_A} \sum_{n=0}^{\infty} \left(\frac{r}{r_A}\right)^n P_n(\cos \psi) \quad (58)$$

Expressing $\cos \psi$ by means of spherical coordinates (35)

θ, λ of A and spherical coordinates θ, λ of the dummy point, from spherical trigonometry we get

$$\cos \psi = \cos \theta_A \cos \theta + \sin \theta_A \sin \theta \cos(\lambda - \lambda_A) \quad (59)$$

It would still be desirable in Eq. (58) to express

$P_n(\cos \psi)$ in terms of the spherical coordinates

θ, λ , and θ_A, λ_A of which ψ is composed according to Eq. (59). This is achieved by the decomposition

formula

$$P_n(\cos \psi) = P_n(\cos \theta_A) + P_n(\cos \theta) + 2 \sum_{m=1}^n \frac{(n-m)!}{(n+m)!} [Y_m^c(\theta_A, \lambda_A) Y_m^c(\theta, \lambda) + Y_m^s(\theta_A, \lambda_A) Y_m^s(\theta, \lambda)] \quad (60)$$

Substituting this formula back into Eq. (58), one

gets:

$$P_n^{-1}(r_A, r) = \frac{1}{r_A} \sum_{n=0}^{\infty} \left(\frac{r}{r_A}\right)^n P_n(\cos \theta_A) P_n(\cos \theta) + \frac{2}{r_A} \sum_{n=0}^{\infty} \left(\frac{r}{r_A}\right)^n \sum_{m=1}^n [Y_m^c(\theta_A, \lambda_A) Y_m^c(\theta, \lambda) + Y_m^s(\theta_A, \lambda_A) Y_m^s(\theta, \lambda)] \quad (60a)$$

Interchanging the summation with volume integration

with substituting Eq. (60a) into Eq. (57) one gets:

$$Wg(r_A) = \sum_{n=0}^{\infty} \left(\frac{a}{r_A}\right)^n P_n(\cos\theta_A) \left[\int_0^{\beta} \frac{a}{G} \sigma(r') \left(\frac{r'}{a}\right)^n P_n(\cos\theta) dr' + \sum_{n=1}^{\infty} \left(\frac{a}{r_A}\right)^n \left\{ \left[\frac{a}{2G} \frac{(n-m)!}{(n+m)!} \int_0^{\beta} \left(\frac{r'}{a}\right)^n P_m(\cos\theta) \right. \right. \right.$$

$$\left. \left. \left. \cos\theta_A \right] P_m(\cos\theta_A) \cos\theta_A \right] + \left[\frac{a}{2G} \frac{(n-m)!}{(n+m)!} \int_0^{\beta} \left(\frac{r'}{a}\right)^n \right. \right.$$

$$\left. \left. \left. \left(\frac{r'}{a}\right)^n P_m(\cos\theta) \sin\theta_A \right] P_m(\cos\theta_A) \sin\theta_A \right\} \right\} \quad (60b)$$

It can be seen that, the zonal harmonic has been separated from other harmonics. Rewriting Eq. (53)

with separating zonal harmonic one gets

$$Wg(r_A) = \sum_{n=0}^{\infty} \left(\frac{r_A}{a}\right)^n A_{n0} P_n(\cos\theta_A) + \sum_{n=1}^{\infty} \left(\frac{r_A}{a}\right)^n \sum_{m=1}^n A_{nm}$$

$$P_m(\cos\theta_A) \cos\theta_A + P_m P_m(\cos\theta_A) \sin\theta_A \quad (60c)$$

Equating Eq. (60b) and Eq. (60c) term by term, we

finally obtain:

$$A_{n0} = \frac{a}{G} \int_0^{\beta} \left(\frac{r'}{a}\right)^n P_{n0}(\theta) dr'$$

$$\left\{ A_{nm} \right\} = \frac{a}{2G} \frac{(n-m)!}{(n+m)!} \int_0^{\beta} \left(\frac{r'}{a}\right)^n \left\{ P_m^s(\theta, \lambda) \right\} dr'$$

(36-1)

(61)

These are the equations relating the potential

(coefficients, in the units of potential, directly to the density distribution ρ . The inverse problem

i.e., the solution for $\rho(\vec{r}_A)$ as a function of

the potential coefficients, is not possible. However,

at least some indirect insight into the density

distribution can be gained from the potential coefficients

namely, there is a relation between some potential

coefficients and the coordinates of the Earth's center

of mass, and the principle moments as well as products

of inertia. To derive this relation, let us adopt

an arbitrary cartesian system fixed to the Earth.

Transforming the spherical coordinates r, θ, λ of the

dummy point used in Eq. (61) into these cartesian

coordinates $\vec{r}' = (x', y', z')$ with the following relation

$$\begin{cases} x' = r \sin \theta \cos \lambda \\ y' = r \sin \theta \sin \lambda \\ z' = r \cos \theta \end{cases}$$

(62)

one can verify that the first twelve potential coefficients acquire the following form:

$$\begin{aligned} A_{00} &= \frac{G}{a} M & B_{00} &= 0 & A_{10} &= \frac{G}{a} M z'_c & B_{10} &= 0 \\ A_{11} &= \frac{G}{a^2} M x'_c & B_{11} &= \frac{G}{a^2} M y'_c & A_{20} &= \frac{G}{a^3} \left(\frac{I'_{xx} + I'_{yy} - I'_{zz}}{2} \right) & B_{20} &= 0 \\ A_{21} &= \frac{G}{a^3} I'_{xz} & B_{21} &= \frac{G}{a^3} I'_{yz} \end{aligned} \quad (37)$$

$$A_{22} = \frac{G}{4a^3} (I'_y - I'_x)$$

$$\beta_{22} = \frac{G}{2a^3} I'_{xy}$$

(63)

all in the units of potential. Here

$$x'_c = \frac{1}{M} \iiint_B x' \varrho(r') dV \quad y'_c = \frac{1}{M} \iiint_B y' \varrho(r') dV \quad z'_c = \frac{1}{M} \iiint_B z' \varrho(r') dV$$

(64)

are the coordinates of the Earth's center of mass in the adopted cartesian coordinate system.

$$I'_x = \iiint_B (y'^2 + z'^2) \varrho(r') dV \quad I'_y = \iiint_B (x'^2 + z'^2) \varrho(r') dV \quad I'_z = \iiint_B (x'^2 + y'^2) \varrho(r') dV$$

(65)

are the moments of inertia of the Earth with respect to the coordinate axes of the primed coordinate system and

$$I'_{xy} = \iiint_B x' y' \varrho(r') dV \quad I'_{xz} = \iiint_B x' z' \varrho(r') dV \quad I'_{yz} = \iiint_B y' z' \varrho(r') dV$$

(66)

are the products of inertia of the Earth with respect to the same coordinate system. Evidently these lower order potential coefficients show not only the location of the Earth's center of mass but also the orientation and dimensions of the ellipsoid of inertia at the origin of the adopted coordinate

(38)

system. the higher order potential coefficients have a similar but more complex (and thus less useful) dynamic interpretation.

The relations (63) are, therefore, of profound importance when we wish to realize the natural geocentric coordinate system. If the coordinate system is selected so as to make $A_{10} = A_{11} = \beta_{11} = 0$ then the system is geocentric. If, in addition,

$A_{21} = \beta_{21} = \beta_{22} = 0$, then the system becomes coaxial

with the principle ellipsoid of inertia, i.e., it becomes the natural geocentric system. In this case, $I'_x = I_1$; $I'_y = I_2$; $I'_z = I_3$ and we have

$$A_{20} = \frac{G}{a^3} \left(\frac{I_1 + I_2}{2} - I_3 \right); \quad A_{22} = \frac{G}{4a^3} (I_2 - I_1) \quad (67)$$

This new (natural geocentric) system is related to the originally adopted system through the following transformation equations:

$$\vec{r} = R(w_1, w_2, w_3) (\vec{r}' - \vec{r}_c) \quad (68)$$

where $\vec{r}'_c = (x'_c, y'_c, z'_c)$ and $R(w_1, w_2, w_3)$ is the rotation matrix. The three rotation angles are obtained by the eigenvalue diagonalization

of the principle tensor of inertia (this is evaluated for the center of mass of the Earth)

$$\begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} \rightarrow \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

tensor of inertia

As already seen, the potential coefficients are all in the physical units of potential, e.g. $\text{cm}^2 \cdot \text{s}^{-2}$. It

is often convenient to work with unitless potential coefficients. For this purpose, the gravitational

potential (53) is often written as

$$w_g(r, \theta, \lambda) = \frac{GM}{r} \left[1 - \sum_{n=1}^{\infty} \sum_{m=0}^n \left(\frac{a}{r} \right)^n (J_{nm} \cos m\lambda + \right.$$

$$\left. + K_{nm} \sin m\lambda) P_{nm}(\cos \theta) \right] \quad (69)$$

where, evidently, $J_{00} = -\frac{GM}{a}$ and $K_{00} = -\frac{GM}{a}$.

Note that when all the potential coefficients are neglected, one gets the spherical approximation of the gravitational potential

$$w_g^s(r) = \frac{GM}{r}$$

which corresponds to $g_g(r) = \frac{GM}{r^2}$. observations have shown that $d_{20} = 1082.63 \times 10^6$, much larger than

(70)

the rest of the potential coefficients which are at most, of the order of 10^{-6} . It will be shown that the $J_2 = J_{20}$ term reflects the ellipticity of the Earth, while the rest of the coefficients reflect the remaining irregularities. In the normal coordinate system we have

$$J_{20} = J_2 = \frac{M a^2}{(-\frac{1}{2}(I_1 + I_2) + I_3)} = \frac{M a^2}{I_3 - I_1} \quad (71)$$

Hence the elliptical approximation of the gravitational potential reads

$$W_g^E(r, \theta) = \frac{GM}{r} - \frac{r^3}{G(I_3 - I_1)} P_2(\cos \theta) = \frac{GM}{r} \left(1 + \frac{I_3 - I_1}{2Mr^2} \right) - \frac{3G(I_3 - I_1)}{2r^3} \cos^2 \theta \quad (72)$$

so far, our attention has been directed to the development of the gravitational potential into spherical harmonic. A very similar argument can be put forward to arrive at the development of W_g into ellipsoidal harmonic functions. The result is:

$$W_g(u, \Theta, \lambda) = \sum_{n=0}^{\infty} \sum_{m=0}^n q_{nm}(u, E, b) (A_{nm} Y_{nm}^s(\Theta, \lambda) + B_{nm} Y_{nm}^s(\Theta, \lambda)) \quad (73)$$

This Equation is valid for the outside of a boundary Ellipsoid (E, b) where $E = \sqrt{a^2 - b^2} = a e$, $\Theta =$ complement of reduced latitude

where $q_m(u, b) = \frac{q_m(\lambda u/E)}{q_m(\lambda b/E)}$, $\lambda = \sqrt{-1}$, and

Q_m are Legendre functions of the second kind

Ellipsoid coordinates u, θ, λ are introduced in

Heiskanen and Moritz, 1967, page 39-45.

To finish this part, let us mention that an

attempt to set up the boundary value problem for w_g has not yet been made. The case dealt with so far was purely hypothetical, since we have no means of

providing the values of gravitational potential on either the boundary sphere or the boundary Ellipsoid.

Model gravity field :

In a variety of tasks, it is advantageous to work with a model gravity field. The degree to which this model field should approximate the actual gravity field depends on the task for which it serves.

gravitational potential (actual)  gravitational potential (model)

It is much more advantageous to work with differences! actual field = model (normal) field + differences

The simplest model is the radial field. This can be

thought of as being generated by either a particle of a negligible size and mass (comparable to that of the Earth, or by a sphere with distribution of masses which would, of course, produce an

identical model field outside the sphere. the potential of such a field is given by Eq. (70), which shows that the field is a function only of the distance from the center of the field. Its equipotential surfaces are concentric, spherical surfaces. A closer approximation to reality is an ellipsoidal model field. In geodesy, It is customary to seek a model field that can be thought of as

a) sharing with the actual field- and thus with the Earth- the spin angular velocity ω

b) being generated by the best-fitting geocentric biaxial ellipsoid defined by a and b

c) having one of its equipotential surfaces - that on which the potential U_0 is equal to the potential W_0 of actual gravity on the geoid -

coincident with the ellipsoidal surface.

of course, such a normal field is obtainable only

to a certain level of approximation that reflects the present level of knowledge about the actual gravity field. Also, it must be remembered that the definition of a normal field neither requires nor stipulates the knowledge of unique mass distribution within the generating ellipsoid. the situation is similar to that of the spherical radial field. we shall now show that the above requirements define a unique normal field. Here, we present the normal potential in spherical coordinates. To obtain the appropriate expression, we begin with the development of the gravitational potential $w_g(r, \theta, \lambda)$ into spherical harmonics

$$w_g(r, \theta, \lambda) = \frac{GM}{r} \left[1 + \sum_{n=1}^{\infty} \left(\frac{a}{r} \right)^n \sum_{m=0}^n (j_{nm} \cos m\lambda + k_{nm} \sin m\lambda) P_{nm}(\cos \theta) \right] \quad (74)$$

and seek its normal part w_g^N . The normal potential $U^S(r, \theta)$ must satisfy the first requirement (a):

$$U^S(r, \theta) = w_g^N(r, \theta) + w_c(r, \theta) \quad (75)$$

The centrifugal potential can be easily formulated by

$$w_c(r, \theta) = \frac{1}{2} \omega^2 r^2 \quad (76)$$

(74)

moreover, to satisfy the requirement (b) the normal

gravity potential must be symmetrical, and thus only the zonal harmonics have to be considered.

It means the gravity potential (normal) is not a

function of variable λ . therefore to now, we can

write:

$$w_g^{(n)}(r, \theta) = G_M \left(1 + \sum_{n=1}^{\infty} \left(\frac{a}{r} \right)^n J_n^{(s)} P_n(\cos \theta) \right) \quad (77)$$

To satisfy the third requirement

$$u^s(r, \theta) \Big|_{r=a} = w_g^{(n)} + w_c = u_0^s = w_0 = \text{constant} \quad (78)$$

we find that all the harmonics of odd degree

($n=2k+1, k=0, 1, \dots$) must disappear. therefore the

final version of gravitational potential (normal) in

spherical coordinates is:

$$w_g^{(n)} = G_M \frac{r}{a} \left[1 + \sum_{n=2, 4, 6, \dots}^{\infty} \left(\frac{a}{r} \right)^n J_n^{(s)} P_n(\cos \theta) \right] \quad (79)$$

This means that the normal field is also symmetrical with respect to the equator. For practical purpose

It is sufficient to take just a few of these normal

potential coefficients. To find the normal gravity

Potential in spherical coordinates $u^s(r, \theta)$, we use the condition (c) which the potential is constant

(us)

on the sphere equal to $U_0^s = w_0 = \text{constant}$.
 the centrifugal potential can be written on
 sphere as:

$$w_c(r, \theta) = \frac{w^2 a^2}{3} (p_{00} - p_{20}) \quad (80)$$

the normal gravitational potential w_g^N can be
 rewritten as (for simplicity in its original form):

$$w_g^N / r_a = \sum_{n=2,4,6,\dots}^{\infty} \int_{(s)} p_{n0} (\cos \theta) + \frac{G M}{r} \rightarrow \int_{(s)} p_{00} \quad (81)$$

Thereafter, the normal gravity potential on the
 sphere $r=a$ can be written as

$$U_0^s = \int_{(s)} p_{00} + \int_{(s)} p_{20} + \sum_{n=4,6,\dots}^{\infty} \int_{(s)} p_{n0} (\cos \theta) + \frac{w^2 a^2}{3} p_{00} - \frac{w^2 a^2}{3} p_{20} \quad (82)$$

$$\text{or } \int_{(s)} p_{00} + \frac{w^2 a^2}{3} p_{00} - \frac{w^2 a^2}{3} p_{20} + \left(\int_{(s)} p_{20} + \left(\int_{(s)} p_{00} + \left(\int_{(s)} p_{20} - \frac{w^2 a^2}{3} \right) p_{20} + \sum_{n=4,6,\dots}^{\infty} \int_{(s)} p_{n0} (\cos \theta) \right) = 0 \quad (83)$$

$$\text{resulting to: } \left\{ \begin{array}{l} \int_{(s)} p_{00} = U_0^s - \frac{w^2 a^2}{3} \\ \int_{(s)} p_{20} = \frac{w^2 a^2}{3} \\ \int_{(s)} p_{n0} = 0 \end{array} \right. \quad n=4,6,\dots \quad (84)$$

$$(85) \quad n=4,6,\dots$$

Finally, we want to express the normal gravity potential $U^S(r, \theta)$ as a function of GM that

can be directly determined, rather than as a function of $U_0^{(S)} = W_0$ that cannot. In other words, an attempt will be made to eliminate W_0

from $U^S(r, \theta)$.

We firstly write

$$U^S(r, \theta) = \sum_{n=0,2}^{\infty} \left(\frac{r}{a}\right)^n \rho_{n0} \rho_{n0}(\cos \theta) + \frac{1}{2} \omega^2 r^2 \quad (85)$$

or using the coefficients of Eq. (84)

$$U^S(r, \theta) = \frac{r}{a} \left(U_0^S - \frac{a^2 \omega^2}{3} \right) + \left(\frac{r}{a} \right)^3 \frac{a^2 \omega^2}{3} - \rho_{20} \quad (86)$$

using the following approximation that

$$W_0^N \approx GM + 0 \left(\frac{1}{r^3} \right) \quad (87)$$

one gets that

$$\frac{r}{a} \left(U_0^S - \frac{a^2 \omega^2}{3} \right) \approx GM \Rightarrow U_0^S = GM + \frac{a^2 \omega^2}{3} \quad (88)$$

and finally

$$U^S(r, \theta) = GM \frac{r}{a} + \frac{r^2 \omega^2}{3} + \frac{r^3}{a^2} \left(\frac{a^2 \omega^2}{3} - r^2 \right) \rho_{20}(\cos \theta) \quad (89)$$

$$U^S(a, \theta) = GM + \frac{a^2 \omega^2}{3} + 0$$

$$r=a$$

(47)

Evidently, the normal potential is defined for every point (r, θ) once GM, w, a are specified. This concludes the proof that the above requirements (a)-(c) uniquely define the normal field.

It can be shown that the normal potential in ellipsoidal coordinates can be obtained from:

$$U^E(u, \theta) = \frac{GM}{E} \operatorname{arctg} \left(\frac{E}{M} \right) + \frac{w^2(u^2 + E^2)}{(1 - p_2(\cos \theta))} + 920(u, E, b) \frac{w^2 a^2}{3} p_2(\cos \theta). \quad (90)$$

Again, the normal gravity potential is function of the size and the shape of the geocentric reference ellipsoid, the Earth's spin velocity and the Earth's gravity-mass constant. Again, the normal gravity potential and field are specified uniquely.

It should be pointed out that for the purpose of defining the normal gravity field, the mass of the Earth is generally considered to include the

mass of atmosphere, i.e., the mass of the Earth is augmented by about $0.89 \times 10^{-6} M$. This is done so that one can regard the normal gravity field generated by the ellipsoid as propagating in empty space above the ellipsoid.

(48)

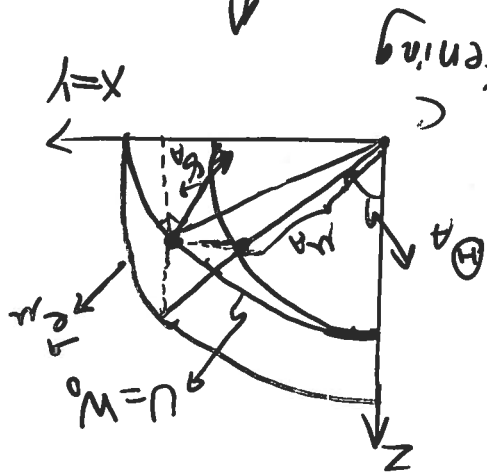
Any model gravity field has model gravity

associated with it. It is, of course, defined as the gradient of the model field potential. In seeking the gradient of the normal gravity potential as given in Eq. (90), the gradient operation is first expressed in the ellipsoidal systems we obtain:

$$\vec{\nabla}_E(u, \Theta) = \nabla U(u, \Theta) = \sqrt{\frac{u^2 + E^2}{u^2 + E^2 \cos^2 \Theta}} \frac{\partial U}{\partial u} \mathbf{e}_u + \frac{1}{\sqrt{u^2 + E^2 \cos^2 \Theta}} \frac{\partial U}{\partial \Theta} \mathbf{e}_\Theta + \frac{1}{\sqrt{u^2 + E^2 \cos^2 \Theta}} \frac{\partial U}{\partial \lambda} \mathbf{e}_\lambda$$

The third term on the right-hand side must disappear since U is not a function of λ .

Also, in the vicinity of the geocentric reference ellipsoid, normal gravity is directed approximately along



the direction of u when the flattening of the ellipsoid is small, as can be seen in above fig. Nowhere is the deviation of these two directions greater than some 13 minutes of arc. To a high degree of accuracy (better than 0.2 μ gal) even the second term containing the rate of change with Θ can be

dropped, and we can write, for the magnitude of normal gravity (the negative sign is because of the opposite direction of $\vec{e}_M$ and $\vec{g}$):

$$(92) \quad g(M, \Theta) \doteq - \sqrt{\frac{a^2 + E^2}{M^2 + E^2 \cos^2 \Theta}} \frac{\partial U}{\partial M}$$

Evaluating now the partial derivative of U from (90),

we obtain:

$$(93) \quad g(M, \Theta) \doteq - \sqrt{\frac{M^2 + E^2}{M^2 + E^2 \cos^2 \Theta}} \left[\frac{GM}{a^2 w^3} - \frac{M^4}{M^2 + E^2} p_2(\cos \Theta) + \frac{2w^2 M}{3} (1 - p_2(\cos \Theta)) \right]$$

when working with normal gravity, it is customary to first evaluate it on the geocentric reference ellipsoid, and then correct for the effect of the location above the ellipsoid. Hence, our first task is to work out a formula for normal gravity on the surface of the ellipsoid, i.e., for g_0 on the surface of Ellipsoid.

g_0 can be evaluated from Equation (93) after

some elementary mathematical manipulations ($M=b$)



$$g_0(\Theta) = g(b, \Theta) \doteq \frac{GM}{a \sqrt{a^2 \cos^2 \Theta + b^2 \sin^2 \Theta}} \left[1 - \frac{3}{2} m + \left(\frac{a^2}{b^2} m + \frac{3}{2} m \right) p_2(\cos \Theta) \right]$$

where $m = \frac{w^2 a^3}{GM}$ is the geodetic parameter.

(94)

From Eq. (94), we can derive, by elementary operations

Clairaut's theorem:

$$f = \frac{a-b}{a} = \frac{2}{5} \frac{w^2 a}{g_E} - \frac{g_E}{g_E - g_P}$$

Here, g_E is normal gravity on the equator, g_P is normal gravity on the poles, and $\frac{g_P - g_E}{g_E} = f$ is known as gravity flattening and $f = \frac{a-b}{a}$ is geometrical polar flattening. The Clairaut's theorem describes the relation between gravity and the size and shape of the Earth.

the Somigliana formula is

$$g_0(\theta) = \frac{a g_P \cos^2 \theta + b g_E \sin^2 \theta}{\sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta}} \quad (96)$$

the second Somigliana formula uses geodetic latitude instead of the second Ellipsoidal coordinate θ :

$$g_0(\varphi) = \frac{a g_E \cos^2 \varphi + b g_P \sin^2 \varphi}{\sqrt{a^2 \cos^2 \varphi + b^2 \sin^2 \varphi}} \quad (97)$$

To transform this equation into the shape Cassini,

$\cos \varphi$ is first expressed as $1 - \sin^2 \varphi$, and geometrical as well as gravity flattening are used. After a few steps, we get

(51)

(52)

(101)

$$+ 0.0000232718 \sin^4 \varphi + 0.00000012625 \sin^6 \varphi \text{ Gal}$$

$$g_0(\varphi) = 978.0327(1 + 0.0052790414 \sin^2 \varphi +$$

IAAG 1980) has an accuracy of 0.7 mGal.

1967. The newer International gravity formula 1980

this formula is called the international gravity formula

with the maximum error of 4 mGal [IAAG, 1971]. Gal

$$g_0 = 978.03185 (1 + 0.005278895 \sin^2 \varphi + 0.000023462 \sin^4 \varphi)$$

(100)

order approximations may be found as:

to the order of ϵ^2 , i.e., to about 50 mGal. Higher

$\beta = f(f - \bar{f})/4$. This formula is accurate only

where $\alpha = \bar{f}$ is the gravity flattening and

(99)

$$g_0(\varphi) = g_E (1 + \alpha \sin^2 \varphi + \beta \sin^2 \varphi)$$

the general expression

Developing the denominator into a power series and neglecting higher order terms, we finally arrive at

(98)

$$g_0(\varphi) = g_E \frac{1 + (f - \bar{f}) \sin^2 \varphi}{\sqrt{1 + (f^2 - 2f) \sin^2 \varphi}}$$

Anomalous gravity field

Disturbing potential

The main application of the normal gravity field is in obtaining anomalous or disturbing potential T which is defined as:

$$T(\vec{r}_A) = W(\vec{r}_A) - U(\vec{r}_A)$$

(102)

This means that our challenge to find $W(\vec{r}_A)$ directly can be converted to finding $T(\vec{r}_A)$ and then knowing $U(\vec{r}_A)$ to derive $W(\vec{r}_A)$ indirectly through $W(\vec{r}_A) = T(\vec{r}_A) + U(\vec{r}_A)$. This is because we have not been able to provide the values of the

$W_g(\vec{r}_A)$ as the boundary values on the boundary

sphere or ellipsoid to solve the Laplace boundary value problem for $W_g(\vec{r}_A)$. Also, the quantity T depicts the regional and local irregularities of W .

Since U models the bulk of the actual gravity

field W , the disturbing potential is much smaller

than either of the other two, therefore, any

approximation used in evaluating T is far less

critical than approximations used in evaluating the other two potentials. Because of the definition of the normal gravity field, the disturbing potential satisfies the Laplace equation outside the Earth. This can easily be shown by substituting for U in Eq. (102) from Eq. (75) and splitting W into gravitational and centrifugal potentials:

$$T(\vec{r}_A) = W_g(\vec{r}_A) + W_c(\vec{r}_A) - [W_g^N(\vec{r}_A) + W_c(\vec{r}_A)] \quad (103)$$

Neglecting the atmosphere, one gets:

$$\nabla^2 T(\vec{r}_A) = 0 \quad (104)$$

everywhere outside the Earth. It is convenient

to seek the disturbing potential in spherical

harmonics. substituting in Eq. (103) from Eqs.

(74) and (79), we obtain: (using minus form of Eqs (74) and

$$T(r, \theta, \lambda) = \frac{GM}{r} \sum_{n=2,4,6}^{\infty} \left(\frac{a}{r}\right)^n \left(\int_n - \int_n\right) P_n(\cos \theta) - \frac{GM}{r} \sum_{n=1,3,\dots}^{\infty} \left(\frac{a}{r}\right)^n \int_n P_n(\cos \theta) - \frac{GM}{r} \sum_{n=1}^{\infty} \left(\frac{a}{r}\right)^n \sum_{m=1}^n \left(\int_m \cos m\lambda + k_m \sin m\lambda\right) P_n(\cos \theta) \quad (105)$$

This Equation is often written as:

$$(106) \quad T(r, \theta, \lambda) = \sum_{n=1}^{\infty} T_n(r, \theta, \lambda)$$

where T_n are regarded as being the components of T for the appropriate orders n , e.g., $J_{5m} \neq J_{5m}$ in Eq. (105)

$$T_5(r, \theta, \lambda) = -\frac{GM}{r} \sum_{m=0}^5 \left(\frac{r}{a} \right)^m (J_{5m} \cos m\lambda + K_{5m} \sin m\lambda) P_{5m}(\cos \theta)$$

Note that for the properly selected normal field

$J_2^{(s)}$ is equal to J_2 . Also if the spherical coordinate system coincides with the natural geocentric system, some potential coefficients disappear and we get,

in particular,

$$T(r, \theta, \lambda) = \frac{GMa^2}{r^3} J_2 \cos 2\lambda P_2(\cos \theta) + \frac{GM}{r} \sum_{n=4,6,8,\dots}^{\infty} \left(\frac{r}{a} \right)^n (J_n^{(s)} - J_n) P_n(\cos \theta) - \frac{GM}{r} \sum_{n=3,5,\dots}^{\infty} \left(\frac{r}{a} \right)^n J_n P_n(\cos \theta) - \frac{GM}{r} \sum_{n=3}^{\infty} \left(\frac{r}{a} \right)^n \sum_{m=1}^n (J_{nm} \cos m\lambda + K_{nm} \sin m\lambda) P_{nm}(\cos \theta)$$

where (Heiskanen and Moritz, 1967)

$$(107) \quad \left. \frac{dJ_n}{dn} \right|_{n=2,3,\dots} = (-1)^{n+1} \frac{3e^{2n}}{3e^{2n}} \frac{(2n+1)(2n+3)}{(1-n+5n \frac{dz}{dz} e^2)}$$

where for Geodetic Reference System 1980 (GRS80) $J_2 = 0.108263 \times 10^{-2}$ and $e^2 = 0.006694380023$ (55)

Gal

Equation (108) can also be written as:

$$(110) \quad T(r, \theta, \lambda) = \sum_{n=2}^{\infty} T_n(r, \theta, \lambda)$$

It should be noted here that we have not still

provided values of the disturbing potential on the boundary (Boundary values of the first kind) or values of the normal derivative of disturbing potential on the boundary (Boundary values of the second kind) or values of linear combination of disturbing

potential and normal derivative of disturbing

potential on the boundary (Boundary values of

the third kind), for the evaluation of coefficients

of J_m and K_m . We will do it in the coming

pages.

It is interesting at this stage to also have a look at what happens to the disturbing potential if the mass of the geocentric ellipsoid generating the normal field - denoted here by M^N - is assessed

improperly. In this case Equations (106) and (110)

(111) will have an additional absolute term:

$$\delta T = -\delta U = T_0 = \frac{r}{GM} - \frac{r}{GM^N} = -\frac{r}{G} \left(\frac{1}{M} - \frac{1}{M^N} \right) = -\frac{r}{G\delta M} \quad (56)$$

considering the accuracy of the value of G_M

better than $10^{-6} G_M$, the corresponding errors

would cause a systematic (constant) global

error in gravity anomalies of the order of 1 mGal

at most. comparable relative errors (of the

error of 10^{-6}) in other parameters of the normal

field, i.e., in w, a and $\frac{J_2}{5}$, have a much

smaller effect on T , as it can be seen in

Equation (105).

Gravity anomaly, the deflection of the vertical,

the geoidal height (three basic gravity field parameters)

the gravity anomaly is defined as the scalar whose value is equal to the difference between the magnitude of the actual gravity on the geoid, g_0 , and the normal gravity on the geocentric ellipsoid, g_0 :

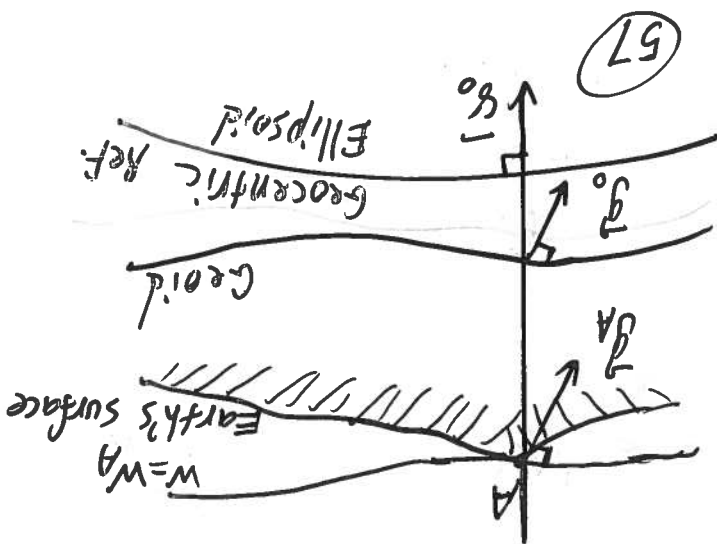
$$\Delta g = g_0 - g_0 \quad (112)$$

when Δg is defined in

this way, it is called

the gravity anomaly on the geoid, or just $U = W_0$

the geoidal gravity anomaly. (57)



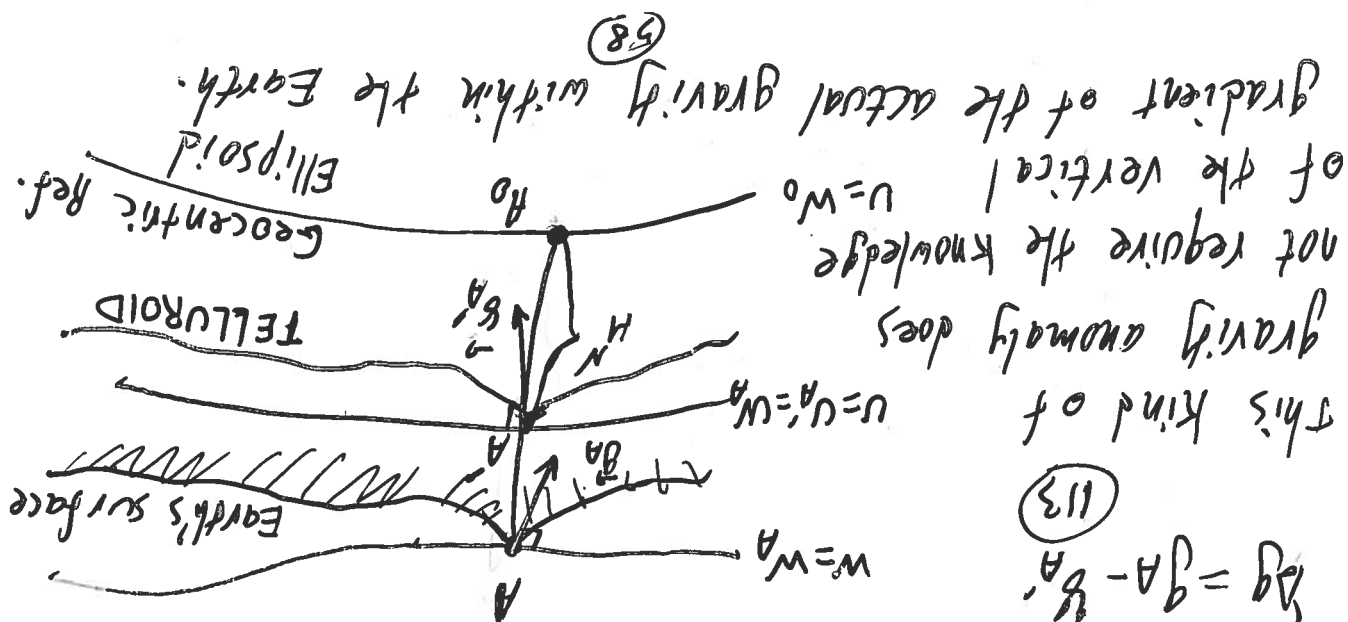
It is evident that on land, g_0 has to be generally deduced from the value g_A observed on the Earth's surface. Analogically, gravity observed on the sea bottom has to be converted to g_0 by applying a suitable correction. There are different ways for the vertical gravity gradients to be valid between the Earth's surface and the geoid. These lead to different reduction procedures, and hence different gravity anomalies, e.g. Free-air gravity

anomaly. A counterpart to the geoidal gravity anomaly is the surface gravity anomaly Δg , which is defined as the difference between the magnitude of the observed gravity taken on the Earth's surface and the normal gravity taken on the telluroid

$$\Delta g = g_A - \gamma_A \quad (113)$$

This kind of gravity anomaly does

not require the knowledge of the vertical $u = w_0$



gradient of the actual gravity within the Earth. (58)

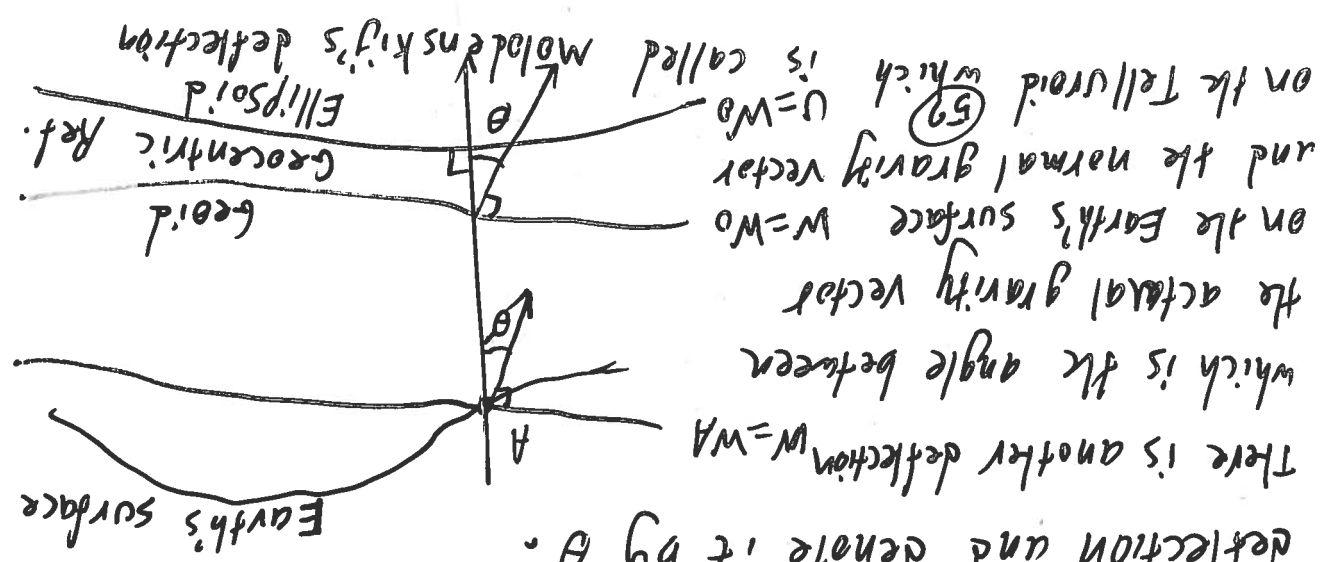
the exact value of the normal gravity on the telluroid needed for this anomaly can be obtained from, e.g.: (Bowie and Avers, 1914)

$$\gamma = 980.624 (1 - 0.002644 \cos^2 \varphi + 0.000007 \cos^2 2\varphi) - 0.3086 H_N - 0.0002 H_N \cos 2\varphi + 7.1 \times 10^{-8} H_N^2 \quad \text{Gal} \quad (114)$$

H_N is called normal height, and is in meters.

A similar situation exists with the deflection of the vertical. the geoid deflection of the vertical is the spatial angle between the normal gravity vector on the reference ellipsoid and the actual gravity vector on the geoid (θ). It can also be interpreted as the maximum slope of the geoid with respect to the reference ellipsoid at the point of interest.

If the angle is reckoned between the normal gravity vector on the reference ellipsoid and the actual gravity vector on the Earth's surface, we speak about the surface deflection and denote it by θ' .



separation between the geocentric reference ellipsoid and the geoid is called the geoidal height, or geoidal undulation and generally is denoted by N . this is sometimes called absolute geoidal height because it relates the geoid to an absolute, i.e., a geocentric reference ellipsoid. there also exists relative geoidal height, which refers the geoid to another kind of reference ellipsoid that is not geocentrically located. At the same time, the quasigeoidal height, called the height anomaly, also has to be mentioned. In the context of gravity field investigations, the quasigeoid (which is not an equipotential surface and does not have any direct relation to gravity) is often treated as an approximation to the geoid. conversion of disturbing potential into other field parameters

Let us now have a look at how these quantities are related to the Earth's gravity potential and in particular to the disturbing potential. Assuming the vertical gradient of U in the figure of next page to be constant between the ellipsoid and the geoid.

It can be seen immediately that the following equation holds :

$$U_B - U_{B_0} = U_B - W_0 = \frac{\partial U}{\partial n} \Big|_{B_0}^N = -g_0 N \quad (115)$$

Realizing that $W_0 - U_B$ is merely the disturbing potential T at B , we can write :

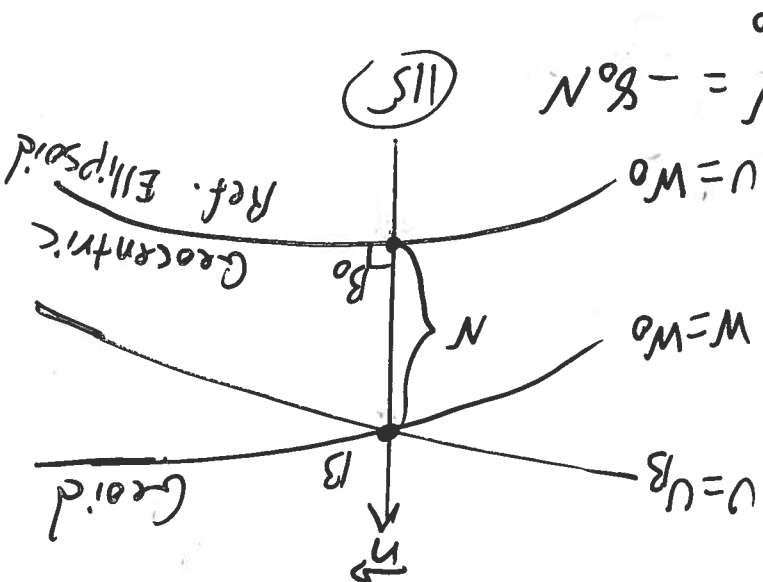
$$N = \frac{T}{g_0} \quad (116)$$

It is understood that T is taken on the geoid. This formula is known as Brun's formula and is used extensively in geodesy. This formula relates the geometric characteristics of the Earth (N) to the physical characteristics (T).

In the above Brun's formula was assumed that the normal gravity field had been properly defined so that the normal potential U on the ellipsoid was equal to the actual potential on the geoid. As it

was pointed out earlier, an error δm in the accepted mass of the Earth cannot be ruled out. An error

δm in M^N would cause not only the disturbing potential T to be in error by δT but also g_0



and a simple calculation convinces us that a relative error of 10^{-6} in the assessment of GM would show as a constant error of $18N \pm 6$ metres. This must then be interpreted as an error in the size of the implied geocentric reference ellipsoid. This is another illustration of how physical and geometrical

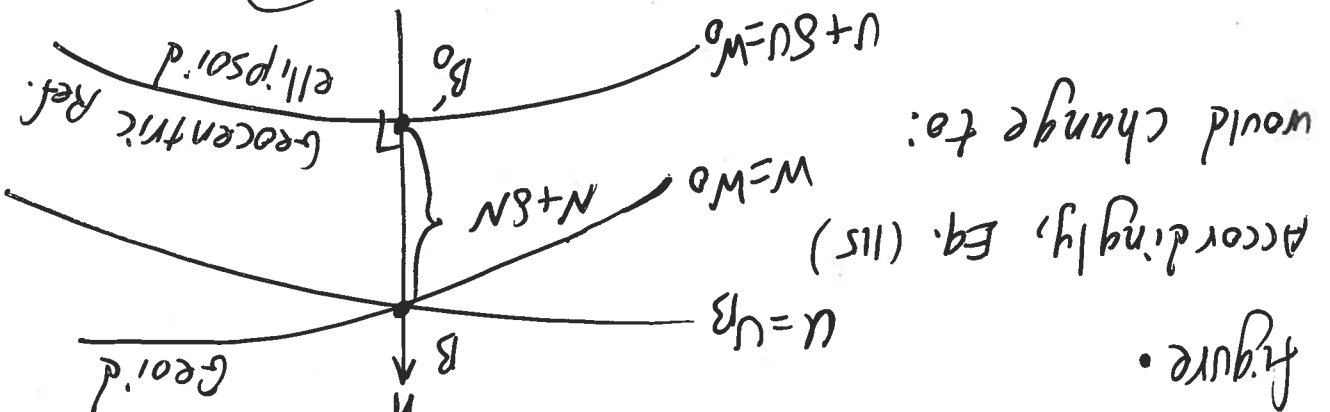
Here

$$\delta N \div \frac{\delta_0}{\delta T} (1 + \frac{N}{T}) \div \frac{\delta_0}{\delta T} \quad (120)$$

$$N + \delta N = \frac{T + \delta T}{\delta_0 + \delta \delta_0} \quad (119)$$

and the generalized Bruns formula would read

$$U_B - (W_0 - \delta U) = -(\delta_0 + \delta \delta_0)(N + \delta N) \quad (118)$$



Then the situation would change from that shown in the figure of previous page to the following figure.

$$\delta \delta_0 = -\frac{G \delta M}{\delta U} = -\frac{r^2}{\delta T} = -\frac{r}{\delta T} \quad (117)$$

would be in error by

clearly, a formula

Brun's can be derived

Consulting with this figure →

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Dg to T, and it requires a slightly more involved

the direction of the ellipsoidal normal $\vec{n}$. The

(63)

where both derivatives can be evaluated on the geoid. This formula is known as the fundamental gravimetric equation of geodesy. If M'' is not accurately assessed, the above formula reads, to a good enough approximation:

$$(125) \quad g_0 - g_B = \delta g \doteq -\frac{\partial T}{\partial H} + \frac{1}{\frac{\partial g}{\partial H}} T$$

of Brun's formula, we finally get:

Adding Eqs. (123) and (124), and making use

$$(124) \quad \frac{\partial T}{\partial H} \Big|_B \doteq -g_0 + g_B$$

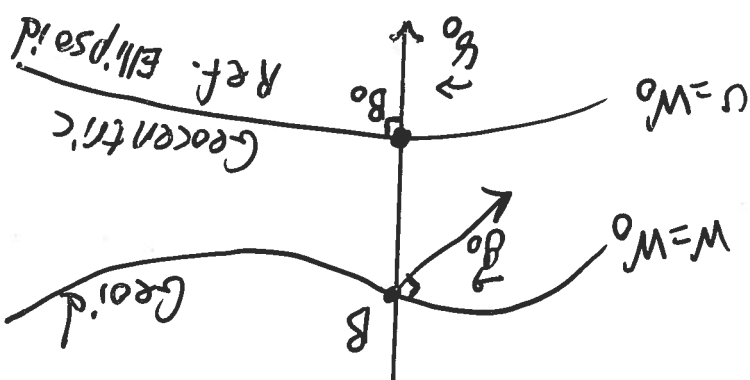
to H. we obtain:

and again derivative is evaluated with respect

$$T_B = W_B - U_B$$

considering

$$(123) \quad -g_B + g_0 \doteq -\frac{\partial g}{\partial H} \Big|_N$$



is equal to zero, one obtains

when it is realized that N is not a function of H and thus its derivative along the normal

$$\Delta g - \Delta g \doteq -\frac{\partial T}{\partial n} + \frac{\partial g}{\partial n} (T + \Delta T) \quad (126)$$

A completely analogous formula holds for the surface gravity anomaly, i.e. :

$$\Delta g \doteq -\frac{\partial T}{\partial n} + \frac{\partial g}{\partial n} T \quad (127)$$

with the only difference being that the derivatives and g are evaluated on the telluroid while T is taken on the surface of the Earth. Also, the normal for the derivatives is taken with respect to the normal equipotential surface passing through A (see Figure of page 63.)

Fundamental gravitational equation of geodesy is a very important formula. It seems that we

have found the boundary values of formula which we were looking for. consider the Laplace Equation of $\nabla^2 T(r) = 0$ for the disturbing potential T .

To solve this formula, we need boundary values (one of the three kinds) to find the coefficients of Δm and Δm (or $\Delta m, \Delta m$) of expansions

of T to spherical harmonics. The fundamental

Equation of below

$$\Delta g = -\frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial}{\partial r} T \quad (128)$$

is obviously the third kind of boundary value

problem, a combination of disturbing potential and

its normal derivative. This will help us to solve

the Laplace Equation for T : After finding T

and knowing already about the normal field

using $W(\vec{r}) = T(\vec{r}) + U(\vec{r})$, we easily find

the gravity potential which we were looking

for. Also, using Bruns's formula $N = \frac{T}{g_0}$,

we can easily find the geoidal height which

it was our another goal in this course.

Finally, the deflection of the vertical θ as a

function of the Earth's gravity potential can be

evaluated. The relation between the geoidal height

and the geoidal deflection components can be

used, since the deflection components at the

point of interest can be considered as slopes

of the geoid with respect to the reference ellipsoid, in the meridian and prime vertical

directions we have

$$\eta = - \frac{dN}{d\sigma} \quad ; \quad \eta = - \frac{d\sigma}{dN} \quad (129)$$

where

$$d\sigma = R d\varphi \quad ; \quad d\lambda = R \cos\varphi d\lambda \quad (130)$$

and finally

$$\eta = - \frac{1}{R} \frac{\partial N}{\partial \varphi} \quad ; \quad \eta = - \frac{1}{R \cos\varphi} \frac{\partial N}{\partial \lambda} \quad (131)$$

where R is the mean radius of the Earth.

the negative signs are due to the sign convention

for deflection components as normally used in geodesy. Now Bruns's formula can be used and

realizing that ξ_0 is not a function of λ , one gets:

$$\xi = \frac{1}{T} \frac{\partial \xi_0}{\partial \varphi} - \frac{1}{R \xi_0} \frac{\partial \varphi}{\partial T} - \frac{1}{R \xi_0 \cos\varphi} \frac{\partial \lambda}{\partial T} \quad (132)$$

It should be intuitively obvious that a constant error δM affecting T has no effect on the deflection components (67)

